Loom: A Distributed Expert-Cache Architecture for Frontier Mixture-of-Experts Models

Technical design and preliminary validation

Author: Parthasarathy Ramanujam **Draft v0.3, 2026-08-01 Status:** design complete for v1; v0 single-node streaming done; v1 first cuts validated on a real three-machine swarm (sharded serving, network-id isolation, RAG gossip, churn repair); Metal and Vulkan GPU backends measured on real silicon; devnet (GLM-4.5-Air) standing up. Implementation progress is tracked in ROADMAP. The wire protocol is specified in SPEC. A dated design-decision log for contributors is in Appendix A.

Contents

Abstract

The strongest open-weight language models are now Mixture-of-Experts (MoE) architectures: Kimi K2 [3], DeepSeek V3 [2], and the GLM family [4], with hundreds of billions to trillions of parameters. At usable quantization their routed experts alone run to hundreds of gigabytes. They do not fit on any commodity machine, and running them today means renting datacenter GPUs that price out individuals and small teams. Multi-machine alternatives exist, but the prevailing approach splits the computation across nodes and ships activations between them every layer of every token. That places the network on the serial critical path, so a slow or lost peer stalls the whole pipeline. The problem Loom addresses is how to serve a frontier MoE model that no single affordable machine can hold, across a pool of ordinary machines on ordinary networks, without putting the network in the critical path of every token.

The opportunity is structural. An MoE model activates only a few percent of its weights per token, and those weights form an immutable, content-addressable corpus: a sparsely accessed blob store rather than a monolithic tensor. Loom keeps computation node-local and moves only weights. The corpus is sharded into content-addressed expert blocks, replicated across coordinated committees of nodes, verified by Merkle digests at every hop, and paged into the token loop on demand. Because only immutable weights cross the network, a lost peer costs a refetch from a replica rather than a broken pipeline.

What has been demonstrated. The inference engines produce correct output on reference checkpoints: the tinyllamas models, and DeepSeek-V2-Lite, a 15.7-billion-parameter MoE, at Q4_K_M quantization. A node holding

33% of that model's shards produced the expected, token-identical completion while fetching the missing experts from a single LAN peer during inference, with every block digest-verified before use. This is a correctness-and-integration result on localhost with scalar CPU kernels (about 0.3 tokens/sec on one core), not a throughput or wide-area result (Section 10).

What is targeted, and not yet demonstrated. The design goal is serving GLM-5.2-class models (744 billion parameters, about 19,200 experts) on swarms of 16 to 32 GB machines. That model has not been run, kernels are not yet vectorized, and there is no batching. One caveat governs expectations throughout (Section 9): on ordinary gigabit Ethernet, distribution buys capacity, meaning the model fits where it otherwise could not, but not speed. Competitive throughput additionally requires 10-gigabit-class fabric, a pinned hot set, and batched serving.

Glossary

Term	Meaning
Expert	One routed feed-forward sub-network (gate/up/down matrices) in one MoE layer. GLM 5.2 has 256 per layer; 8 activate per token.
Expert shard	The unit of distribution and compute: one expert's weights, about 19 MB at int4. A fetched expert shard is exactly what a MoE matmul consumes.
Resident chunk	A roughly 16 MB piece of the resident bundle, the non-expert weights (attention, embeddings, shared experts, router, norms) that every full node holds in full.
Manifest / version id	The list of shard digests plus layout. Its Merkle root is the model's version identity, which nodes pin requests to.
Full node	Holds the resident bundle and a subset of expert shards, runs local inference, serves shards to peers. The supply side.
Light node	Holds no weights and runs no engine; exposes the same local API and delegates to full nodes. The demand side.
Committee	A group of full nodes that collectively holds the complete shard set, a self-sufficient serving cell.
MLA	Multi-head Latent Attention [1]: attention with a low-rank compressed key/value cache (about 576 values per token per layer), used by the target models.
Mesh	The peer table each node builds from gossip announcements, used for discovery and fallback routing.

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1. Introduction

Loom targets operators who want to run frontier open-weight MoE models on hardware they already have: a homelab, a research group's workstations, or a small trusted cluster, rather than rented datacenter GPUs. It is not built for interactive single-user chat on a cold model over a slow link (Section 9 explains why), and v1 assumes a cooperative, operator-run swarm rather than an open adversarial network (Section 7).

Three facts frame the problem.

1. **The best open models are MoE.** Kimi K2 (about 1T total, 32B active), DeepSeek V3 (671B, 37B), and the GLM family activate roughly 3 to 5 percent of their parameters per token [2][3][4].
2. **The weights do not fit cheaply.** At int4, the routed experts of a GLM-5.2-class model total about 370 GB, beyond any commodity machine and beyond most workstations.
3. **The activated set is small and skewed.** A single token reads on the order of 600 expert blocks (about 11.4 GB for GLM 5.2), drawn Zipf-like from a fixed corpus of roughly 19,200. This is a sparsely accessed, immutable, content-addressable blob store.

Loom's premise follows. This is a distributed storage and caching problem in the shape of an inference problem. The corpus is stored once across a swarm, replicated by policy, cryptographically verified, and paged into computation on demand, the way a distributed filesystem treats cold blocks rather than the way a tensor-parallel runtime treats a partitioned weight matrix.

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2. Related work

Systems that run large models across multiple machines can be organized by their unit of distribution and the failure and trust properties that follow from it.

System	Unit distributed	What crosses the network per token	Failure model	Commodity-NIC friendly
Petals [5]	Transformer layers	Activations, every layer	Serial: a slow or lost stage stalls the token	Partially (activations are small but on the critical path)
exo [6]	Layer partitions	Activations	Serial pipeline	Partially
FlexGen [9]	Offloaded tensors (single node)	None (host/device)	N/A (not distributed)	N/A
PowerInfer [8]	Hot/cold neurons (single node)	None (GPU/CPU split)	N/A	N/A
llama.cpp [7]	None (single node) or layer-split	Activations if split	Serial if split	Yes single-node
Loom	**Expert block (weights)**	**Expert blocks only** (cacheable, immutable)	**Soft: re-fetch from a replica**	**Yes for capacity; speed needs 10 GbE**

Loom reuses the GGUF weight format and GGML quantization layouts [7] and the DeepSeek MLA and sparse-routing model shape [1][2]. It introduces three things: the expert block as the unit of distribution, committee-based redundancy with coverage by construction, and a token-loop fetch that doubles as replication. Its closest philosophical relative is content-addressed storage (Merkle-verified immutable blocks [10]) applied to model weights rather than files.

Layer-split systems such as Petals and exo are deployed and effective. The distinction Loom draws is structural, not a claim that they do not work. Because they ship activations, which are mutable, per-request, and order-dependent, the network sits on the serial critical path. Loom ships only immutable weights, so a lost peer costs a re-fetch rather than a broken pipeline.

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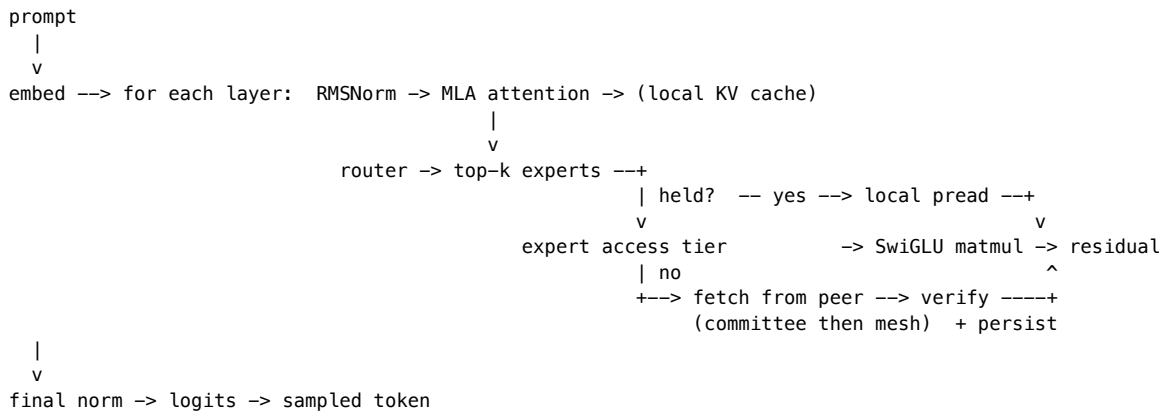
3. Core thesis: experts flow, not activations

Every full node runs the complete forward pass locally. The only data that crosses the network during inference is an expert block: immutable, content-addressed, cacheable, and identical for every user of the model. Three properties follow.

- **Soft failure.** A lost peer costs a re-fetch from a replica, not a broken pipeline. A lone node still functions, streaming experts from its own disk.
- **Compounding cache.** Frequently used experts come to reside on many nodes, so the network is touched only on the cold tail of the access distribution.
- **KV stays home.** The target models use MLA [1], which compresses the key/value cache to about 576 values per token per layer. Distributing that mutable, latency-critical, per-session state would put it on the network for a negligible capacity gain, so Loom does not.

The governing caveat, developed in Section 9: on gigabit Ethernet the swarm buys capacity, not speed. Loom is a serving-throughput architecture first and an interactive-latency one second.

Figure 1. Inference data path (one full node). Only routed-expert reads may leave the machine; everything else is local.



4. System architecture

Loom is a single binary written in Zig. It separates a data plane (weights at rest and in flight, and the local forward pass) from a control plane (membership, gossip, repair, metering). Two inference engines share a common set of quantized-matmul kernels.

- The **MLA** engine (`deepseek2`), covering the DeepSeek/Kimi family: MLA attention, YaRN context extension [11], and a byte-level BPE tokenizer read from GGUF metadata. Validated on real DeepSeek-V2-Lite weights (Q4_K_M).
- The **GQA** engine, covering `llama` (including Mixtral), `qwen2moe`, `qwen3moe` and `glm4moe` in one forward pass. These architectures differ only in optional pieces over a shared skeleton, so the engine detects them from the tensors a file contains rather than from per-architecture assumptions: QKV biases, per-head Q/K normalization, dense or mixture-of-experts FFN, an optionally sigmoid-gated shared expert, leading dense layers, a post-attention norm in place of `ffn_norm`, and trailing MTP blocks that are skipped. Only the rotary-embedding style is pinned per architecture, because an incorrect choice yields fluent but wrong output rather than an error. The dense llama path is validated against the tinyllamas reference models at F32, Q4_0 and Q8_0.

Both engines share one router (sigmoid or softmax gating, selection bias, top-k with optionally renormalized and scaled gates, shared experts), so a new architecture requires an attention variant rather than a new engine. The distribution plane was never architecture-specific: expert sharding keys on the GGUF tensor-naming convention that every mixture-of-experts conversion follows.

Weights remain in their GGML quantized formats in a read-only memory map, and each matmul is a fused, vectorized kernel over the raw bytes, distributed across a worker pool by row, so no tensor is dequantized wholesale. Two families are supported. *Affine* formats (F32/F16/Q4_0/Q5_0/Q8_0/Q4_K/Q5_K/Q6_K) reconstruct a value arithmetically. *Codebook* formats (IQ1_S, IQ1_M, IQ2_XXS, IQ2_XS, IQ2_S, IQ3_XXS, IQ3_S, IQ4_NL, IQ4_XS, and MXFP4) instead store an index into a static grid table, so a transcription error in that table would silently corrupt weights rather than fail; every codebook decoder is therefore checked bit-for-bit against the reference implementation's output on stored vectors. The source layout appears in Appendix B.

5. Expert-aligned sharding

The unit of distribution equals the unit of computation. An expert shard is one expert's gate/up/down matrices in one layer, described as a byte-extent list over the GGUF file's three-dimensional expert tensors. A fetched expert shard is precisely what a MoE matmul consumes: no reassembly, no erasure coding, no partial reads on the token path.

Everything that is not a routed expert (attention, norms, shared experts, embeddings, router weights, the file header) forms the resident bundle, split into roughly 16 MB resident chunks that every full node holds in full.

This is what keeps compute node-local. Any node can run the dense path and the router by itself; only routed-expert reads may leave the machine.

Every byte of the file belongs to exactly one shard, either an expert shard or a resident chunk. Each shard carries a SHA-256 digest, and the Merkle root over the digest list, together with the layout, is the model's version identity, which nodes gossip and pin their requests to. Integrity is therefore free at every hop: a shard from any source is checked against the local manifest before it reaches disk or a matmul.

Sizing for the GLM-5.2-class target:

Quantity	Value
Expert shards	19,200 (75 MoE layers x 256 experts)
Expert-shard size	about 19 MB (int4)
Routed corpus	about 370 GB, stored once across the swarm
Resident bundle	about 10 GB, on every full node
Logical file	about 380 GB; each node's copy is sparse (about 60 GB of real bytes at defaults)
Manifest / holdings bitmap	about 1 MB / 2.4 KB

Measured on real DeepSeek-V2-Lite Q4_K_M (10.4 GB): 1,664 expert shards of 4.98 to 6.02 MB, plus 73 resident chunks totaling 0.78 GB; manifest 204 KB; holdings bitmap 218 B.

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6. Distribution protocol

This section states the protocol's shape and invariants. Field-level detail is in SPEC.

Component	Role	Key property
Bootnode	Onboards joiners; assigns each a committee and a least-covered-first want-set	Coverage by construction; not in the inference path; trusted in v1
Committee	A group of full nodes that collectively holds every shard	Completeness means at least one holder per shard; filled toward redundancy R (default 2)
Gossip mesh	Every node announces its holdings every 3 s; peers merge into a peer table	Transitive discovery; carries committee membership
Query path	Fetch a missing shard: committee members first, then the mesh	Digest-verified; fails only if no reachable holder exists anywhere
Eager repair	A 2 s loop re-fetches any wanted-but-missing shard from all known peers	Dead peers are retried, not forgotten; a returning holder is drained within one tick
Liveness	First-hand contact only; 30 s TTL gates holder claims, never addresses	Hearsay cannot revive a dead peer, so death is detectable without committee membership
Wire messages	Binary frames (Heartbeat, Announce, ExpertRequest/Response) with adaptive Snappy [13]	Requests pin the version, so a cross-version peer is refused (the hardfork guard)

Two design choices hold the protocol together.

Coverage by construction. Rather than hope random placement covers every shard, the bootnode assigns each joiner the shards its committee currently covers least. Completeness (every shard has a holder) and redundancy (R holders) are distinct thresholds a committee crosses as it fills. When all committees are saturated at R, the next joiner opens a new one.

Membership is gossip-derived, not static. Announcements carry committee ids, so earlier members learn later joiners automatically and the committee view survives the bootnode's departure. Heartbeats (5 s) carry liveness

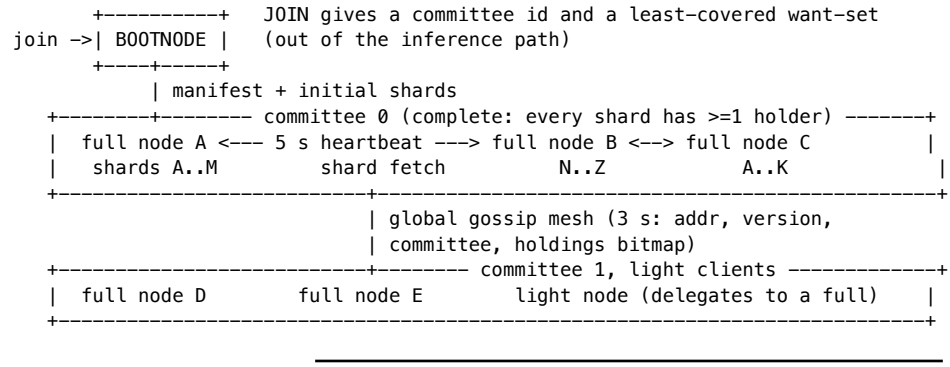
plus the manifest version, a monotonic holdings sequence number, a bitmap digest, and a load hint. That is enough to detect staleness and spread load without shipping the full bitmap on every beat.

Liveness is first-hand only, and it gates holder claims rather than addresses. Two properties are easily conflated and must not be. An address is useful indefinitely — the eager-repair policy never forgets one, so a returning holder is picked up without re-bootstrapping. A *holdings claim* is only as good as the evidence that its owner still answers. Loom therefore tags every peer-table update with how it was learned: `first_hand` (we exchanged with that peer, or it connected to us) or `hearsay` (another peer listed it). Only first-hand contact refreshes a peer's last-seen time; a peer unheard-from for 30 s (ten gossip rounds) stops being offered as a holder while its address is retained.

The distinction is not academic. Without it no death is detectable in a swarm of three or more, because the survivors keep echoing the dead node's entry to one another and each echo renews it. Measured on a three-node run, an origin holding the full model was killed and both survivors still advertised it as a live holder six minutes later. Nor can liveness be delegated to the committee heartbeat, which watches committee members only: the origin is the bootnode and sits in no committee, so the node that was the sole holder of 332 of 1,664 expert shards was monitored by nobody.

Under-replication is reported, not inferred. A node's status line names the number of shards fewer than R live peers hold. Redundancy is an arithmetic consequence of membership, not a setting that can be asserted: two joiners at `--hold-fraction 0.40` cover 80% of one copy and cannot reach R=2 however correctly the shards are assigned. Measured on the same run, exactly 332 shards sat at one holder under `--r-target 2`, and before this was surfaced nothing said so.

Figure 2. Committee and mesh.



7. Trust and security model

v1 assumes a cooperative, operator-run swarm and provides cryptographic integrity relative to a trusted manifest.

Assumptions. The bootnode is operated by the swarm operator, peers report their holdings honestly, and the transport is a trusted LAN.

Guarantees. Every shard is checked against its manifest digest before disk or matmul, so corruption, bit-rot, or a wrong-bytes peer is caught at every hop. The manifest's version id binds the shard layout, so a manifest whose extents do not tile the file exactly is rejected on parse. Version pinning on every request isolates model versions and, later, hardforks.

Non-guarantees. Which manifest is trusted is not free. Content addressing secures integrity within a manifest, not the choice of manifest. A node adopts its Merkle root on first contact (trust-on-first-use) or from operator config, and a hostile first contact could pin an alternate, internally consistent but poisoned root. Holdings claims are also unverified: a peer can advertise shards it lacks. This is caught reactively, when a fetch returns wrong or absent bytes and the requester falls through to the next peer, so committee completeness is a construction-time property under honest participation, not a cryptographic invariant. v1 does not defend against availability attacks (service refusal, mesh poisoning with fake holders, request flooding).

v2 (designed, not built): untrusted peers. The planned untrusted-peer layer adds rateless (RLNC) wide-area propagation gated behind homomorphic-hash pollution defenses [14], redundant recompute with M-of-N voting on sampled tokens, and TEE attestation. Erasure and network coding are confined to the propagation and durability planes; they are never permitted on the token path, where only direct fetch of an original block is allowed.

Known limitations of the current implementation are consolidated in Section 11.

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8. Node classes and compensation

Loom defines two node classes.

Full nodes hold the resident bundle and their assigned expert shards, join committees, serve shards to peers, and run inference. They are the supply side. The "compute node-local, only expert blocks cross the network" property of Section 3 applies to the full-node inference data path; request, response, and control traffic naturally also cross the network.

Light nodes hold no weights, no store, and no engine, a footprint of a few megabytes suitable for a low-memory device. A light node exposes the same local API a full node does, so applications are unaware of the distinction, and delegates each request to a full node with round-robin failover across its configured backends. It stamps its own client identity on every request, and a caller-supplied identity is dropped. Light nodes are thin clients, not swarm participants: they contribute no storage or serving, so capacity planning must size the full-node fleet for aggregate light-node demand.

Compensation. Full nodes are meant to be compensated by light nodes for the inference they serve. v1 implements the accounting mechanics and leaves settlement as an integration seam. Each full node keeps a per-client ledger, where allowance equals free quota plus purchased credits minus token usage. A metered response carries its cost and the remaining balance, and an exhausted client is refused with `payment_required` before any compute. Credits are added through an admin-gated `credit` operation whose proof field a real payment rail would verify. This is scoped as a trusted-operator accounting demonstration, not a settlement system. Its limitations are enumerated in Section 11.

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9. Deployment and performance model

Held shards are a disk budget served by `pread`. RAM carries only the compute working set.

Resource	Minimum	Recommended
RAM	16 GB (about 10 GB mmap'd resident bundle, 0.7 GB MLA KV, 2 to 3 GB expert cache)	24 to 32 GB
Disk	75 GB NVMe (resident bundle plus about 50 GB expert shards)	150 GB NVMe
Experts held	about 1,300 (25 GB)	about 2,600 (50 GB), 13.7% of GLM 5.2
Network	1 GbE (capacity only)	10 GbE+ (also latency)

The 16 GB minimum is conditional on the resident bundle. A per-tensor-class breakdown from the real converted GLM 5.2 file is required before it is asserted as a default rather than a target. For DeepSeek-V2-Lite the measured resident bundle is 0.78 GB.

Swarm sizing against redundancy R over the 370 GB corpus (R = 3 target, R = 2 floor): at 50 GB per node, completeness (R = 1) needs at least 8 nodes, R = 2 needs at least 15, and R = 3 needs at least 22. Eight nodes at 100 GB reach R = 2.

Performance model (analytical, not yet measured). Per-token time is about `t_compute + misses * t_fetch`, where `misses` is the number of a token's routed experts absent from the local store and page cache, and `t_fetch` is about `RTT + shard_size / bandwidth` per miss. Fetches within a layer parallelize, so the layer cost is the maximum, not the sum, over its 8 or fewer misses. A cold 19 MB miss costs roughly 160 ms on 1 GbE and 16 ms on

10 GbE before protocol overhead. The consequence is direct. With a cold working set on gigabit Ethernet a single stream is seconds per token, unusable interactively. Serving becomes viable insofar as pinning, the Zipfian access skew, and organic replication drive the steady-state miss rate toward zero, and batching amortizes what remains. The `t_compute` term is itself scalar today; hardware-tailored backends (CPU SIMD, and GPU via Metal/Vulkan/CUDA, staged behind one interface in roadmap issues #10 to #14) are the lever that reduces it. No network-tier throughput measurements exist yet; Section 10 lists them as the primary measurement agenda.

Positioning follows from the model: pool commodity machines so a frontier MoE fits where it otherwise could not; speed is a separate problem, bound by network and pinning. The differentiators against layer-split systems are structural (weights not activations, soft failure, verified weights, self-healing membership). The differentiator against single-box streaming is aggregate capacity.

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10. Preliminary results

These are early results, a case study plus live single-machine and localhost multi-node runs, not a throughput evaluation. The evidence and its boundaries:

Claim	Configuration	Evidence	Boundary
Engine correctness	DeepSeek-V2-Lite Q4_K_M; tinyllamas F32/Q4_0/Q8_0	Coherent, expected completions on real weights (validates kernels, MLA, RoPE convention, K-quants, BPE, YaRN)	Not a factual-accuracy benchmark; single-sequence; scalar kernels
Expert-aligned sharding	Real 10.4 GB model	Round-trips to 1,664 expert shards plus 73 resident chunks; layout partition property-tested	Fully covered
Partial-store inference	33% store (573 of 1,737 shards), localhost, one peer, scalar	Token-identical to a full-copy run while fetching 641 experts (3.5 GB, about 29 ms/fetch, 0 failures); grew to 1,214 shards held	Not NIC throughput or failover under WAN-like RTT and loss
Committee and repair behavior	Live multi-node runs	Observed: bootstrap, gossip discovery, committee formation and saturation, heartbeat death detection, eager repair, mesh fallback	Not adversarial availability
Weight-integrity defenses	Live	A corrupted on-disk shard is caught on store-open, cleared, and re-fetched; malicious non-partition manifests rejected on parse	Fully covered
Metering	Live, two full nodes and one light node	Transparent delegation with cost and balance; per-provider ledgers; deterministic `payment_required` on	Trusted-operator only (Section 11)

Claim	Configuration	Evidence	Boundary
		exhaustion; resume after credit	
Target frontier deployment	GLM-5.2-class	Analytical sizing only	Not run

The abstract's headline, the "Paris." completion, should be read as expected, token-identical output under a partial store, demonstrating the fetch-verify-persist loop and engine correctness, not as a factual-accuracy evaluation.

Measurement agenda (not yet done). Tokens per second as a function of miss rate and NIC tier; repair time after killing a sole holder; the committee-saturation join sequence at scale; and end-to-end throughput once hardware-tailored kernels land.

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11. Limitations

Consolidated so a reader sees the full picture in one place.

Compute and coverage.

- Kernels are scalar (no SIMD or GPU), about 0.3 tokens/sec on a 15.7B model on one core.
- Single-sequence only; no continuous batching.
- GLM 5.2 itself has not been run; the deepseek2 path is the architectural proxy.
- No wide-area or contended-network measurements exist.

Trust (v1 is operator-trusted).

- Manifest choice is trust-on-first-use; a hostile bootstrap can pin a poisoned root.
- Holdings claims are unverified; committee completeness assumes honesty.
- No transport encryption or authentication (plaintext TCP), which is LAN-appropriate, not public.
- Availability attacks (service refusal, mesh poisoning, flooding) are not defended.

Metering (trusted-operator accounting demo).

- Client ids are self-asserted strings, so rotating an id resets the free quota.
- Ledgers are per-provider, so round-robin across N full nodes yields N times the free quota.
- `credit` is admin-gated but its payment proof is unverified in v1.
- No signed receipts, so neither party can prove the other's ledger wrong.
- Prerequisites for real compensation: cryptographic client identity, payment-proof verification, signed usage receipts.

Operational.

- The distributed engine is exercised via `loom gguf run`, not yet the node's RPC path.
- Persisted organic replicas have no disk cap or eviction yet, so a hot node's disk grows.
- ENR advertising and hardfork coordination are designed but unimplemented.

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12. Roadmap

Milestones with exit criteria; full tracking in ROADMAP.

1. **Hardware-tailored kernels** (issues #10 to #14). Exit: a measured tokens/sec improvement over scalar on DeepSeek-V2-Lite on a single node, via a backend interface with a scalar differential oracle; CPU SIMD first, then GPU.
2. **Serve the distributed engine through node RPC.** Done: `loom node` serves the distributed GGUF (deepseek2) engine over its RPC and OpenAI surfaces, answering inference requests via the token-loop peer fetch, not only `loom gguf run`. A partial-store node's output is byte-identical to a full-store node's for the same prompt.

3. **Network-tier evaluation.** Exit: a published table of tokens/sec vs miss rate across 1 and 10 GbE, and repair time after a sole-holder failure.
4. **Hardfork coordination.** Exit: a majority of nodes adopting a new manifest version, with the existing version guard enforcing isolation during transition.
5. **v2 trust layer.** Exit: untrusted-peer serving with pollution-resistant propagation and sampled compute verification.

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13. Conclusion

Loom reframes frontier-MoE inference as a distributed-storage problem: store the expert corpus once across a swarm, verify it cryptographically, and page it into node-local computation on demand. The consequences (soft failure, a compounding cache, and self-healing committee membership) follow from moving immutable weights rather than mutable activations. The implementation demonstrates the core loop end-to-end on real weights under a partial store, and is explicit about the boundary: today's result is correctness and integration on localhost with scalar kernels, and the pitch is capacity, not speed. Pooling ordinary machines lets a frontier MoE model fit where it otherwise could not. Making it fast is a separate problem, bound by network and kernels, and the roadmap treats it as such.

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Appendix A: Decision log

For contributors. This append-only log records design decisions with dates and rationale. It is maintained alongside the code and is not part of the paper's narrative. Superseded decisions are struck through, not deleted. Where this log, the spec, and the roadmap differ, the spec governs the p2p protocol and the roadmap governs status.

Date	Decision	Rationale
2026-07-11	Distribution unit is the expert, not the layer; compute node-local; never distribute KV	MoE sparsity plus MLA's tiny KV; avoids activations in the serial network path
2026-07-11	Content-addressed shards plus a Merkle-rooted manifest	Free integrity and poisoning check at every hop; root doubles as version id
2026-07-11	No coding (EC or RLNC) in the token loop, ever	Decode-before-use adds latency where it cannot be hidden; coding stays in propagation and durability
2026-07-12	Target toolchain: Zig 0.16.0, matching the sibling zeam project	Shared toolchain plus reusable deps (ENR, SSZ, snappy)
2026-07-12	v1 direction: GGUF interchange format; ENR, gossip, and request-response p2p; majority hardforks; boot-time peer sync	Requirements of record; supersedes an earlier Hyperswarm/Hyperbee sketch
2026-07-12	Weight advertising: both ENR (compact summary within the 300 B limit, a bitmap digest plus seq) and a global gossip topic (full bitmap)	ENR for discovery-time selection; gossip for live freshness
2026-07-13	Churn repair is maximally eager; an always-on loop retries all known peers and dead peers are retried, not forgotten	Chosen over lazy repair-on-miss
2026-07-13	Gossip is epidemic announce-and-exchange over the peer table; transport swappable for gossipsub later	Table semantics are transport-independent
2026-07-19	deepseek2 rope is NORM (adjacent-pair), not NEOX	Empirical: DeepSeek's (d/2,2) transpose before rotate_half nets to adjacent-pair on the stored layout; NEOX degenerates
2026-07-19	Response payloads carry no digest; the requester verifies against its own manifest	The manifest is the only trust root
2026-07-19	Shard equals one expert block plus mandatory 16 MB resident chunks; about 19 MB per expert gives 19,200 shards for GLM 5.2	Shard unit must equal the matmul's consumption unit; metadata stays trivial
2026-07-19	Bootnode assigns committees least-covered-first toward R = 3 target, R = 2 floor; saturated committees spawn new ones	Coverage by construction; bootnode out of the inference path
2026-07-19	Fetched shards are persisted, not evicted, so fetch-on-demand is organic heat replication	Disk is the cheap resource; hot experts gain holders by being used
2026-07-20	Wire messages v1: binary frames, adaptive snappy; heartbeat carries seq, digest, and load but not the bitmap	Quantized payloads are incompressible; heartbeats stay small; staleness detected cheaply
2026-07-20	Committee membership is gossip-derived, since announces carry committee id	Fixes static-at-join membership; survives bootnode death
2026-07-20	Two node classes: light nodes (no weights or engine; delegate) and full nodes (shards plus inference)	Local inference for low-memory devices without weakening supply
2026-07-20	Compensation: per-client metering ledger on each full node; `payment_required` enforcement; `credit` op as the settlement seam	Accounting precedes payment rails; ledgers are per-provider

Date	Decision	Rationale
2026-07-20	Redundancy R = 3 target, R = 2 floor (default 2); completeness means R >= 1	Completeness and redundancy are distinct thresholds
2026-07-20	Placement is least-covered-first committee assignment; random `--hold-fraction` is the no-bootnode fallback	One live placement policy
2026-07-20	Binary frames are normative; the line protocol is legacy and debug during migration	Prevent forked implementations
2026-07-20	Hardware-tailored compute backends behind one `Backend` seam (scalar oracle; CPU SIMD then GPU); planned as issues #10 to #14	Per-node throughput gates the distribution story
2026-07-20	Code-audit hardening: version root binds layout; hot-path and serve re-verify; store-open re-audits held shards; resident completeness gate; publish-after-verify cache; credit admin-gated; light node forces client id; token clamp; account cap; snappy decode cap; bounded peer table; monotonic holdings seq; connection caps; death triggers wanted re-replication	Close the gap between "verified before use" and the code; ship the availability bounds the spec named
2026-07-20	Documents restructured to standard whitepaper format (front matter, TOC, glossary, related work, references, claim-to-evidence table, limitations); terminology split (expert shard vs resident chunk); decision log demoted to a contributor appendix	Editorial reviews: read as a whitepaper, scope claims precisely, cite external work
2026-07-20	Add an OpenAI-compatible HTTP API as the north-facing client surface (`--openai-port`); client identity moves to `Authorization: Bearer`; MCP is not the serving path (a node is more naturally an MCP client)	Ecosystem clients work with no adapter; bearer identity is out-of-band and not prompt-forgeable, unlike the native RPC `client` field. Shipped as a skeleton (`src/node/openai.zig`): transport, routing, shapes, identity; generation/streaming TODO
2026-07-21	A `Generator` abstraction (`src/node/generator.zig`) unifies the loom-format and distributed GGUF (deepseek2) engines; both RPC and OpenAI serve through it; the node auto-serves the GGUF engine when an expert-sharded store is attached and its resident bundle is complete	Serves the distributed engine (token-loop peer fetch) through the node, not only `loom gguf run` (roadmap item 2). Store mutation from serving-fetch and eager repair serialize on one engine mutex
2026-07-21	Light nodes delegate the OpenAI surface too (`src/node/light_openai.zig`), a metered reverse proxy to full-node OpenAI endpoints; the light node forces its client id via `Authorization: Bearer` (caller bearer dropped)	OpenAI clients can point at a low-memory light node; identity is forced so a light node cannot be an open proxy, parallel to the native-RPC rule
2026-07-22	SSE streaming for `stream:true` (`src/node/openai.zig`): OpenAI `text/event-stream`, one chunk per token then `[DONE]`, over both engines. Backed by an optional per-	Chat clients that default to streaming work; client disconnect aborts generation, metering still charged

Date	Decision	Rationale
	token callback on `engine.generate` and a `generator.TokenSink`	
2026-07-22	Per-model chat templates (`src/gguf/chat_template.zig`): detect the format from GGUF `tokenizer.chat_template` (or arch), render `messages[]` per format (deepseek/chatml/llama2/llama3/gemma/mistral/generic), `--chat-format` override. Format detection plus built-in renderers, not a Jinja engine	Real chat models need their own prompt format. DeepSeek-V2 (the validated target) is faithful
2026-07-22	Special-token-aware tokenization (`src/gguf/special.zig`): both BPE and SPM splice control (type 3) / user-defined (type 4) tokens to atomic ids, longest-match with a first-byte filter, splitting the input before normal encoding	Chat markers (chatml/llama3/gemma) and control tokens tokenize exactly instead of being split. No change when no special appears (no regression). Caveats: parse_special always on (injection); SPM dummy-prefix on the first segment only
2026-07-24	Parse-special injection hardening: `parse_special` threaded through the tokenizer chain; off for raw prompts and text-marker chat formats (deepseek/llama2/mistral), on only for special-marker scaffolds (chatml/llama3/gemma)	Untrusted input can no longer inject a control token on the raw-prompt or RPC path, nor via text-marker chat content (the validated DeepSeek-V2 target is fully safe). Segment-encoding content for special-marker formats is the remaining follow-up
2026-07-27	Codebook quantizations implemented (`src/gguf/iq.zig`): IQ1_S/M, IQ2_XXS/XS/S, IQ3_XXS/S, IQ4_NL/XS and MXFP4, with grid and sign tables transcribed mechanically from llama.cpp (`src/gguf/iq_tables.zig`)	Most modern GGUF repositories publish mostly IQ variants, so the affine-only kernel set excluded the majority of available checkpoints. MXFP4 additionally covers microscaling checkpoints. Verified bit-for-bit against llama.cpp's own `dequantize_row_*` on stored vectors (`src/gguf/iq_vectors.zig`), because a wrong codebook entry corrupts weights silently instead of failing
2026-07-27	MoE routing extracted to `src/gguf/moe.zig` and shared by both engines, including the expert-to-shard binding used for distributed fetch	The routing was never DeepSeek-specific: upstream funnels every MoE architecture through one routing function differing in four knobs (gating function, selection bias, gate renormalization, constant scale). Sharing it means a new architecture needs an attention variant, not a new engine
2026-07-27	The llama engine generalized into one GQA engine covering `llama` (Mixtral), `qwen2moe`, `qwen3moe` and `glm4moe`; optional features detected from the file's tensors, with only the RoPE style pinned per architecture	Distributed serving previously required `deepseek2`, even though sharding, sync and repair were already architecture-agnostic — a Mixtral store distributed correctly but could not be served. Feature detection follows the checkpoint rather than a table of beliefs about each architecture, which is also what makes the qwen3 explicit `head_dim` and the glm4 NextN skip fall out naturally. Verified end to end: for all five architectures a node holding roughly 30% of shards serves with hit rate below 1 and

Date	Decision	Rationale
		produces output token-identical to a full-copy origin
2026-07-28	A node serves a GGUF locally when it is not expert-sharded, instead of falling back to the synthetic checkpoint	A dense model has no routed experts, so it shards into fixed ranges and cannot be served *distributed* — but it is a complete model file, and answering from an unrelated synthetic checkpoint reads as a broken model rather than as an unsupported topology
2026-07-29	The 1.9x whole-layer result is validated on TinyLlama-1.1B only; Mistral-7B on the same machine is memory-bound and tests nothing. The device KV cache is released when calibration declines every reader	Mistral-7B runs at 0.27 tok/s here with the process at 116% CPU and 2.75 GB resident against a 4.37 GB model -- stalling on page faults. Calibration measured 3442 ms/tok against 3414 and correctly declined, since a ~262 us command buffer is invisible against 3.4 seconds. So the GPU result holds for cache-resident tensors (TinyLlama's are 6.5 MB) and is unproven at DRAM-bound sizes, which is also where ZINC's kernel advantage disappears. This is the same constraint the distributed thesis exists for, arriving from the other direction: a 16 GB box cannot hold a 7B model and its caches, let alone a large MoE
2026-07-29	The recorded whole-layer path is the default on a GPU machine, chosen by timing a real token both ways at load (25% margin, `--no-gpu-layers` to force the host path)	The unit of measurement has to be the whole token: nothing smaller is honest about submission cost, because timing one operation in a tight loop lets successive commits pipeline while the engine serialises them. The per-operation calibration this replaces reported the fused FFN block at 1.107 ms against a CPU 9.837 ms, when the same CPU block measured 0.489 ms elsewhere, and acting on it took decode from 56 to 9.1 tok/s. The whole-token measurement predicts 119 tok/s where 109 is observed. A build with no GPU backend never enables it: `layerBlock` declines, both timings match, and the margin is not met
2026-07-29	A frame abstraction on the compute seam (`beginFrame`/`layerBlock`/`endFrame`) records a whole GQA layer per command buffer; `--gpu-layers` measures 110.3 tok/s against 57.7 on the CPU path, with correct output	The seam's one-shot entry points each created a command buffer, committed, waited and copied back, so a token cost ~150 submissions at ~262 us regardless of kernel quality -- and the same kernels behind a recorded path are 1.9x faster. ZINC's Metal on the same model measures 52.9-56.1. Recording introduces failure modes that immediate submission does not, all of which produce fluent, wrong output: host writes to a shared scratch buffer are no longer ordered against the recorded reads (twenty-two layers read the last layer's norm weights); and a dispatch grid must follow each kernel's

Date	Decision	Rationale
		rows-per-SIMD-group, since Q4_K takes two rows per group and Q6_K one, so dispatching Q6_K with the Q4_K count silently computes half the output -- TinyLlama's `ffn_down` is Q6_K. A unit test passed throughout because it ran only `pos=0`, where RoPE is the identity and single-position softmax makes attention a passthrough: the fixture had switched off both things most likely to be wrong
2026-07-29	The Metal Q4_K matvec is at the machine's DRAM bandwidth ceiling and level with ZINC's kernel; the remaining GPU gap is command-buffer submission, not kernel quality	Measured amortized (many dispatches per command buffer, as a kernel microbenchmark must be), loom reaches 114-116 GB/s from 32k rows up against a ~110 GB/s streaming-read ceiling and ZINC's reported 109 GB/s at 27 MB. An earlier reading of ZINC's numbers as ~3x faster compared loom's per-dispatch timings against ZINC's amortized ones -- ZINC reports 259.81 us for a shape where one command buffer alone costs ~262 us, so its figures cannot include one. The same kernel takes 0.504 ms per-dispatch and 0.323 ms amortized at 32k rows: ~180 us of submission, which is the whole difference between losing to the CPU and beating it 1.3x. ZINC remains genuinely ahead at cache-resident shapes (239 GB/s at 2.25 MB vs 67), where matvecs are latency-bound and its register-level work pays; adopting three of those techniques changed nothing at DRAM-bound sizes, consistent with there being no headroom there
2026-07-29	Metal gains Q6_K dmmv, a batched Q4_K gemm for prefill, and fused grouped-query attention over a device-resident KV cache. All are validated against exact CPU references and all are **disabled by default**, behind `--gpu-ops`	Each kernel is correct and each loses. The reason is not kernel quality but that calibration, which times one operation in a tight loop, systematically overestimates the GPU: consecutive submissions pipeline there and are serialised in the engine. Acting on that verdict took decode from 56 to 9.1 tok/s -- a 6x regression chosen by a measurement that looked rigorous. Two further traps were found and fixed: a device KV mirror written by decode but not by batched prefill left zeros at every prefilled position, and attention over them produced fluent, wrong text; and maintaining that mirror when nothing reads it cost 12,672 device writes over a 576-token prefill, taking decode to 3 tok/s with the GPU otherwise unused. The verdict is still computed and printed, because seeing it is useful; it is not obeyed until calibration measures a whole layer as the engine issues it

Date	Decision	Rationale
2026-07-29	Metal is built by default on macOS; the CPU/GPU choice per operation is measured on the loaded model's own tensors at startup (25% margin, tie resolves to CPU) rather than compiled in; the fused FFN block is exported through the compute seam	A shipped binary that is CPU-only cannot use a GPU backend however good it is, and a compiled-in row threshold is a measurement taken on someone else's machine -- the crossover is a ratio between a given box's GPU and its CPU cores. Measured against ZINC (Metal) on TinyLlama-1.1B Q4_K_M on an M5: loom 55.7-56.2 tok/s decode, ZINC 52.9-56.1 -- loom matching a Metal engine while running entirely on the CPU, because calibration found the GPU slower at every shape this model issues. This refutes the inference that equal throughput implies equal GPU optimisation, and bounds the remaining GPU prize: on unified memory the measured GPU win is ~1.5x above 65k rows and nothing below it, so the payoff is in large models and prefill, not small-model decode
2026-07-29	The GGUF serving path gains a local RAM tier (LRU of verified expert blocks, sized by the largest *expert* shard rather than the largest shard); MLA attention absorbs W_k into q and accumulates values in compressed space; Q4_1/Q5_1 gain int8-activation kernels; a `LOOM_PROFILE=1` per-phase decode profiler; optional zero-copy weight serving from a read-only mapping (`--mmap-weights`, off by default)	Decode was 1.07 tok/s and the reasons were all outside the kernels. Expert `get` was 54-58% of a token because every routed expert was re-read from disk *and* re-hashed on every access with nothing cached; MLA was decompressing K and V per head per cached position, O(seq) matvecs where the whole point of MLA is that K and V are never materialised; and `ffn_down_exps` in this Q4_K_M checkpoint is Q5_1, so a third of every expert fell to the exact dequantize path. 584 -> 272 ms/token, 1.07 -> 3.03 tok/s end to end. The profiler is itself a decision: two well-argued theories together moved the number by 20%, and a hypothesis that survives a measurement which should have refuted it means the measurement is what needs fixing
2026-07-29	Measured, on a 16 GB machine serving an 8.9 GB model, single-node decode is bound by getting ~1.16 GB of expert weights per token off disk, not by compute. Four residency strategies (no cache, 2 GB heap, 8 GB heap, zero-copy mmap) all land within 480-362 ms/token and merely move the cost between `get` and the matmul	This is the regime the distributed thesis is *for* -- the whitepaper's own bandwidth table says a single commodity box cannot hold a large MoE's working set -- but it also means single-node numbers on an oversized model measure the storage path, not the engine. Engine-efficiency comparisons against other runtimes need a model whose expert corpus fits in RAM, or the comparison measures page-cache policy
2026-07-28	The Metal Q4_K matvec reads four quant bytes per lane instead of one, with a branchless 6-bit scale unpack; `MIN_GPU_ROWS` drops 100,000 -> 65,536; the GPU scaling benchmark reports best-of rather than mean	The previous "61 GB/s is the unified-memory wall" conclusion was drawn from our own kernel's achieved bandwidth, never from the machine's. A pure streaming-read probe reaches ~110 GB/s on the same M5, so the wall was ours: the kernel was coalesced but quarter-

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		width, with too few loads in flight to cover their latency. Widening alone is a regression below 131k rows because the sub-block index then varies per lane and the scale unpack's branch diverges within the SIMD group; both changes are needed. Result 63 -> 94 GB/s at 131k rows, 1.5-1.6x. Decode on a 1.1B model is still CPU territory -- its largest tensor is 32,000 rows and a command buffer costs ~262 us fixed. Means were replaced by best-of because GPU contention with the window server moved repeated measurements of an unchanged kernel by 50%, which had briefly made a regression look like a 1.4x win
2026-07-28	Peer liveness is evidence-tagged: only `first_hand` contact refreshes last-seen, `hearsay` (a peer appearing in another peer's table) does not; a peer stale for 30 s is withheld from holder selection and from the peer count while its address is retained for eager repair. Nodes report the number of shards held by fewer than R live peers	Liveness previously came only from the committee heartbeat, which watches committee members; a bootnode/origin belongs to no committee, so the node solely holding 332 of 1,664 expert shards was monitored by nobody. Independently, hearsay refreshed last-seen, so in any swarm of three or more the survivors' echoes kept a dead peer alive indefinitely — measured, an origin killed at t=0 was still advertised as a holder at t+6 min. Separating address (keep forever) from holdings claim (expire) preserves the eager-churn policy while making death detectable. Under-replication is surfaced because R is an arithmetic consequence of membership, not a setting: n nodes at hold-fraction f cannot exceed R = n*f
2026-07-28	A chat UI is compiled into the binary and served on its own listener (`--ui-port`, default 8555), sharing the HTTP implementation and generator with the OpenAI API; plus a periodic console status line (`--status-secs`)	Nothing in the system reported its own state: exercising a node meant hand-writing HTTP, and membership and holdings move with no request arriving, so an event-only log made a churning node look idle. Serving the page same-origin with the API removes any CORS or host configuration. The UI separates decode speed from time-to-first-token, because on a partial node the first token also pays for the prefill's expert fetches
2026-07-28	`/health` reports whether the served weights are one of loom's random-weight fixtures, and the UI says so	A fixture answers with meaningless text by construction; unlabelled, that is indistinguishable from a broken model and was the first thing a new user saw
2026-07-28	Q4_1 and Q5_1 implemented, completing every non-ternary GGML type	Three Q5_1 tensors out of 377 blocked an entire 8.9 GB checkpoint. llama.cpp quant *mixes* fold a handful of legacy types into otherwise K-quant files, so one missing format costs a whole model

Date	Decision	Rationale
2026-07-28	SIMD kernels via `@Vector` plus row-parallel matvec over a worker pool (issue #11)	Inference ran at roughly 1% of the machine's memory bandwidth on one of ten cores: the limit was scalar single-threaded code, not the absence of a GPU. Measured 23.6x on a 10-core M5 (1.1 -> 26.0 tok/s, TinyLlama 1.1B Q4_K_M). Row-splitting is exact rather than approximate, since each output row is an independent dot product with no reassociation, so results stay bit-identical to the serial path and the tests assert it. The pool is scoped to a generation because Zig 0.16 moved condition variables under `Io`, which the kernels deliberately do not depend on, so parked workers would otherwise spin
2026-07-28	int8-quantized activations for the Q4_K, Q6_K and Q8_0 kernels	Profiling showed 76 to 92 percent of a quantized matvec was the dequantize step, not the dot: each block was expanded into a 256-float scratch buffer and pushed through L1 only to be read back. Quantizing the activation once per matvec and dotting against packed weights as integers removes the buffer and uses lanes four times as wide. Measured 1.1 -> 51.9 tok/s overall (47x from the original scalar path). Unlike the earlier steps this is an approximation, about 0.4 percent per activation element, so results are no longer bit-identical to the dequantize path and the tests bound the error against the mass of the summed terms rather than against the heavily cancelled result
2026-07-29	~~A whole MoE layer is recorded as one command buffer (`moeFfnBlock`: zero the accumulator, then every routed expert's gate/up/SwiGLU/down and its weighted add, one submission for the layer), plus a Q5_1 matvec kernel so the DeepSeek checkpoint's `ffn_down_exps` no longer forces the layer back to the host. It is used only when the expert source hands back pointers that stay valid across later fetches -- a mapping, or an LRU with at least as many slots as the layer selects~~	~~Per-expert dispatch cost one ~262 us command buffer per expert, seven per layer, against expert FFN kernels of ~250 us; the missing Q5_1 kernel was worse than its own tensor's cost, because declining one tensor declines the whole layer. The measured outcome is that it changes nothing end to end, and the reason is the more useful finding: on a 16 GB machine serving a 9.4 GB store, `expert get` is 36-40% of a token and varies by more than the entire FFN bucket between processes -- four alternating warmed runs gave 1.31/2.01 tok/s batched against 2.36/1.58 per-expert, ranges that overlap completely. Correctness is established instead of speed: the block matches the CPU expert loop numerically in test, and 48 greedy tokens are identical either way. Calibration on this model still selects the host path for every operation, so the code ships disabled here and is kept for the regime it was built for -- a

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		model whose experts are resident, where fetch stops dominating~~
2026-07-29	The compute seam exports `moeFfnBlock` and `ExpertRef`; the engine's batched MoE path actually runs	`deepseek.zig` guarded the branch with `@hasDecl(backend, "moeFfnBlock")` against the *seam*, which did not re-export it. `@hasDecl` is comptime, so the guard was comptime-true, the whole branch was never analysed, and it compiled and ran as if the feature existed. A capability probe against the wrong module is indistinguishable from a backend that declines
2026-07-29	A weight mapping is handed to the compute backend as device allocations and made resident (`registerArena` / `materializeArenas`, MTLResidencySet), split at the device's `maxBufferLength`. Measured 4.1-5.4 tok/s against 3.5-3.6 for the 8 GB heap cache, page-ins over a generation 6.9 GB -> 1.4-2.2 GB	Two earlier conclusions were wrong and are corrected here. "This machine cannot make a 10.4 GB model resident" -- it can: llama.cpp (build 34ce48d, Metal) on the *same file and machine* maps it into one 9,880 MiB buffer, pins it with a residency set, offloads 28/28 layers and runs at 46.8-58.6 tok/s with 249 MB of page-ins. And "on unified memory the GPU win is ~1.5x above 65k rows and nothing below" was measured on TinyLlama, where the model is 6.5 MB and only kernel speed is in question; on a 10.4 GB MoE the GPU also buys residency. What was *right*: loom's CPU engine is at parity with llama.cpp's (2.52 vs 2.60 tok/s), so the kernels were never the problem. Three failures were needed to get here, each silent: registering into a context the node had not built yet; wrapping without `requestResidency`, which leaves the pages as evictable as any other mapping; and asking for 9.7 GB in one allocation when `maxBufferLength` is about half of RAM. None of them failed loudly, which is why the banner now prints the resident byte count and the reason when there is none
2026-07-29	Two pre-existing defects, both found by the residency work rather than by a test: `Source.get` read a resident chunk into an expert-sized cache slot, and `Store.deinit` never freed the lazily-allocated `verified` bitmap	The first is a heap overflow -- 16 MB into a 6.3 MB slot -- on the resident-gate loop that every node runs at startup, so it had been corrupting memory on every distributed launch since the RAM tier landed. It only ever surfaced under `-Doptimize=ReleaseSafe`; ReleaseFast wrote past the slab and carried on producing coherent text. The lesson is not about the bounds check but about the benchmark: months of numbers were taken on a build whose safety checks were off, on a path where the overflow was guaranteed
2026-07-29	Metal gains Q5_0 and Q8_0 matvec kernels, and a test that asserts `moeFfnBlock` *accepts*	Reading the checkpoint's tensor types rather than trusting the quant mix's name: DeepSeek-

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	every down-projection type real checkpoints use rather than skipping when it declines	V2-Lite Q4_K_M stores `ffn_down_exps` -- one of the three tensors in every routed expert -- as Q5_0 in some layers and Q8_0 in others, and contains no Q5_1 at all. Neither had a pipeline, so `moeFfnBlock` declined at `pipelineFor(e.down.ty)` for every MoE layer, and the batched path was dead on the one model it was written for. The Q5_1 kernel added earlier was for a type this file does not contain. It survived review because the block's existing test treats a decline as `error.SkipZigTest` -- a backend that refuses everything passes a suite built that way -- so the new test fails instead, and was checked by removing the pipeline again
2026-07-29	The backend counts command buffers and the profiler reports submissions per token; the profile is a rolling window rather than a cumulative total	Counted rather than timed, because a count is deterministic and a wall-clock number on a swapping machine is not: a command buffer costs ~262 us fixed whatever it holds, so `count x 262 us` is a floor on the token that no kernel work can move. The measurement then refuted the optimisation it was taken to justify. The MLA engine has no recorded-layer path where the GQA engine does, and the plan was to build one -- but submissions are 26-28 per token, ~6.9 ms of 128, so the whole recorded path was worth at most 5% and was not built. What the count did confirm is that `moeFfnBlock` now runs: 26 submissions across 26 MoE layers is one per layer, where per-expert dispatch would be 500+. Windowed rather than cumulative because the first touch of an expert hashes its whole 6 MB shard to verify it, so a short profile is mostly that transient and a long one averages it away -- neither says what the node is doing now. Watching the windows converge (178.6 -> 157.8 -> 147.4 -> 136.6 -> 129.5 ms/token) separates the two for the first time
2026-07-29	Releases, the container image and `install.sh` build **ReleaseSafe**, not ReleaseFast; CI additionally runs the suite optimized	A node accepts weight shards from peers and writes them into buffers, so the difference between the two modes is whether an out-of-bounds write traps or executes. It was not hypothetical: a cache slot sized for a routed expert was handed a larger resident chunk on the startup path every node runs, and ReleaseFast wrote 16 MB into a 6.3 MB buffer and carried on producing plausible text for however long it had been there. Measured cost, alternating runs on DeepSeek-V2-Lite decode: 80.8/82.6 ms/token ReleaseFast against 91.0/91.8 Releas-

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		eSafe, about 11%, and *none* of it in the GPU kernels -- the expert FFN block is 41.1 ms in both, so what is being paid for is bounds checking on host loops. 11% of throughput is a fair price for a daemon on a network; ReleaseFast remains available and is the honest choice for a kernel benchmark, which is why the docs now say which mode a number came from. The suite runs optimized in CI as well, because Debug and ReleaseSafe differ in what the optimizer may assume, and a bounds check only Debug performs is not one this project ships
2026-07-30	Two-machine run over a real WAN (Mac <-> Hetzner, 25.8 ms RTT, 29 MB/s): a node holding 3.6% of shards produced output token-identical to a full copy, fetching 1,063 expert shards from the peer, each digest-verified. 24 tokens in 429.6 s	The correctness and integration claim holds across machines and across a wide-area link, not just on localhost -- which is what this test existed to check. The throughput is what the bandwidth table in Section 9 predicts and is not the point
2026-07-30	**`--hold-fraction` is not a capacity cap, and the storage half of the thesis is therefore unenforced.** It bounds the bootstrap fetch only; a shard fetched at token time is verified, persisted and marked held, and nothing evicts it	Found by running the capacity test rather than by reading the code: the node above went from 3.6% to 93.1% of the corpus, 1.5 GB to 9.0 GB on disk, in the course of generating 24 tokens. The whitepaper's claim that the swarm stores the corpus *once* -- sharded and hot-replicated instead of copied per node -- does not survive this: every serving node drifts toward a full replica, so N nodes converge on N copies and the capacity benefit that distinguishes Loom from per-node streaming evaporates. Fetch-on-demand as organic heat replication is a deliberate and defensible design (`expert_fetch.zig`), but without an eviction policy it has no upper bound. What is needed is a real resident-set cap with eviction, at which point `--hold-fraction` becomes the knob the documentation already claims it is. Until then the honest statement is that Loom demonstrates distributed *fetch*, not distributed *capacity*
2026-07-30	`--hold-fraction` becomes a real cap: holdings are enforced continuously, the least-recently-used expert is evicted once the count is exceeded, and its blocks are hole-punched back to the filesystem. Resident chunks are never evicted. Measured on the same two-machine setup: holdings pinned at 489/1737 (73 resident + 416 experts, exactly the cap) and stayed flat while serving, where the uncapped node had run to 93.1%	This is what the storage half of the thesis requires: without it every serving node converges on a full replica and N nodes hold N copies. Two things had to be right and only the first was obvious. Clearing the bit makes the cap true, but punching the hole is what makes it true *on the disk* -- and APFS refuses an unaligned punch outright, so the first version freed nothing while the holdings count sat correctly at 28.2% and the store grew 1.8 -> 7.6 GB anyway. Extents start and end at arbitrary offsets, so the range has to be aligned inward to whole

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		blocks, losing at most one block at each end. The test asserts allocated blocks rather than the bitmap, and was checked by reverting the alignment. The cost is now visible: at a 25% cap over a 29 MB/s link the node re-fetches evicted experts and 24 tokens exceeded the 900 s generation deadline, so the cap is a capacity knob with a latency price, not a free win
2026-07-30	Bulk range sync fails over a wide-area link (`EndOfStream`, 0 of 905 assigned shards) while the per-token fetch path over the same socket succeeds	The node still reached 93.1% holdings -- through 1,063 individual token-time fetches instead of one bulk transfer, which is why the run took 430 s where a bulk sync at the measured 29 MB/s would have taken about three minutes and then served fast. Only a real link surfaced it: the same code path is exercised on localhost by the multi-node walkthrough and passes there, so whatever the framing assumption is, it holds at loopback latencies and does not hold at 25.8 ms
2026-07-30	The p2p server refreshes its deadline between requests; `fetchFromPeer` returns partial progress on error; the RAM-budget default is a quarter of physical memory, clamped to [0.5, 8] GB	Fixed and re-verified on the same two machines: bulk sync now reports `synced 905/905 ranges (5275.3 MB)` where it had reported `EndOfStream, 0/905, 0.0 MB`, and the origin runs at 1.92 GB RSS on the 7.7 GB box that previously OOM-killed it at 7.4 GB, with no `--ram-gb` given. The deadline was taken at accept and never refreshed, so a 30 s budget covered a whole connection instead of idle time -- fatal for a ~5.4 GB bootstrap at 29 MB/s, and invisible on loopback where the same sync finishes in seconds. The count was lost separately: `fetchFromPeer` accumulated into a local and returned `!FetchStats`, so an error discarded the tally for everything already written, which is why the node claimed 0 while its holdings bitmap disagreed. Two independent defects presenting as one symptom, and only the second was cosmetic
2026-07-30	The shared expert is timed into the `expert ffn` bucket. Folding it into `moeFfnBlock` was tried, measured 6% slower, and reverted	It is a 2816-wide dense FFN on every layer -- ~225 MB/token, a third of what the routed experts move -- and it sat outside both the block and the profiler's timing, so it landed in "everything else": 30% of a token and the largest thing nobody had explained. Attributing it was free. Batching it was not: A/B'd in one binary, folded runs 97.9/92.1 against 92.8/86.2 ms/token not folded. The fold is correct in principle (one command buffer instead of extra dispatches) and loses anyway, because it moves the work from the CPU onto a Metal matvec

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		that is slower at 1408-2816 rows -- which is the same verdict calibration reaches on its own, and why `--gpu-ops` is needed to force the GPU path at all. The lever is therefore the kernel at those row counts, not the batching around it: at ~27 GB/s against this machine's ~110 GB/s streaming ceiling, the expert FFN is where the remaining time is, and adding work to a losing kernel makes it worse
2026-07-30	Q4_K is two kernels compiled from one source -- four rows per SIMD group below 4096 rows, two above -- selected together with their dispatch grid by a single `dmmvFor`. Two correctness bugs found on the way, one of them pre-existing and silent	The scaling sweep started at 2048 rows and so had never covered the shapes an expert FFN actually issues (1408 routed, 2816 shared, 2048 dim), which is where half a token is spent. Extended down, it showed 33-38 GB/s there against 113 at 131k rows: not bandwidth, since 1408x2048 of Q4_K is 1.6 MB and sits in cache, but activation reuse -- each group re-reads the whole 8 KB vector to serve two rows. Four rows per group takes 1408-row throughput to ~47-53 GB/s and costs ~7% above 5632 rows, so both variants ship and the shape picks. Pairing the pipeline with its grid in one function is not tidiness: the two must agree, nothing else enforced it, and dispatching one kernel with the other's grid silently computes part of the output vector -- the worst bug in this work, twice. The new oracle test then found a third instance immediately: `dispatchThreads` may launch a smaller final threadgroup, in which `simdgroups_per_threadgroup` is not the full count, so `tgid * nsg + sgid` addresses the wrong rows and the tail is never written -- `matvec` then copies stale scratch. Latent since the kernels were written, invisible because every shape tried divided evenly (65,536 at two per group, 1408 at four); 1409 does not. Grids now round to whole threadgroups. The GPU still loses to the CPU at these shapes (0.034 ms against 0.027 at 1408 rows), so calibration's verdict is unchanged and there is no end-to-end gain yet -- the kernel is closer, not ahead
2026-07-30	Kernel variants are compared by an *interleaved* benchmark -- each variant timed round-by-round inside one run -- and the four-rows-per-group Q4_K kernel is 1.8x the two-row one at the expert-FFN shapes (87-94 against 47-53 GB/s at 1408-2048 rows). Threadgroup staging of the activation vector was built, measured consistently worse, and removed	The instrument had to be fixed before any of this could be believed. Comparing variants across separate runs does not work on this machine: the same binary reported 62.4, 52.9, 42.8 and 46.1 GB/s at 2816 rows on four consecutive runs, a +-20% band that swallows the differences being measured and had already produced two confident wrong conclusions. Interleaving the variants within a run puts

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		whatever else the GPU is doing onto all of them equally, and repeatability goes from +-20% to +−3%. The sweep was also measuring the wrong thing outright: it dispatched `cx.q4k` with a `((rows + 1) / 2)` grid left from when Q4_K covered two rows per group, so against the four-row kernel it launched twice the groups needed and timed the waste -- 47 GB/s reported where an honest grid gives 89. Both now go through the same `dmmvFor` the engine uses. The staging idea was well-founded (at 1408 rows the activations are 2.8 MB of traffic against 1.6 MB of weights) and still lost, because the 16 KB threadgroup allocation costs more occupancy than the sharing saves; eight rows per group lost for the same reason at 1408, where 44 threadgroups no longer fill the device
2026-07-30	A MoE layer is recorded as four phases across all experts -- every gate and up, then every SwiGLU, then every down, then one fused weighted reduce -- instead of a per-expert chain. Expert FFN 56.1 -> 49.1 ms/token, token 98.1/81.7 -> 81.8/76.6	The per-expert form put a barrier at every step, about thirty for a layer, and a barrier is a pipeline drain rather than a fence on one buffer -- so at most two dispatches could ever be in flight. Experts are independent until the final sum, so the ordering was never required. Four barriers now, and twelve-way parallelism in the first phase for six experts. The accumulation had to change with it: a `scaled_add` per expert all write the same accumulator and so needed a barrier between each, which one `moe_reduce` over the strided outputs removes entirely. Measured by alternating two binaries in one session, because a cross-run comparison of this size is not resolvable on this machine -- the expert-FFN ranges do not overlap (48.1-50.0 against 53.5-58.6) where the token totals nearly do. Also worth recording: the 87-96 GB/s the interleaved kernel benchmark reports is *cache* bandwidth, since it reuses one 1.6 MB matrix; the engine streams ~1.1 GB per token from DRAM, so the kernel figure is not the engine's ceiling and the gap between them is not kernel quality
2026-07-30	Three kernel-level probes, all of which refuted the hypothesis they were built for, plus a profiler split that found where the expert-FFN time actually is	The bucket was 48-60 ms/token against a MoE block that measures 11.1 ms for 26 layers, so the question was where the other 5x went. Not DRAM: reading 24 distinct 1.6 MB matrices is *faster* than reusing one (86.2 against 75.0 GB/s), so the kernel is not cold-miss bound at these shapes. Not the untuned quantizations: `ffn_down_exps` is Q5_0 or Q8_0 and those kernels run at 121.6 and 106.6 GB/s against

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		Q4_K's 87.5 -- the two written last are the fastest. What it is: the shared expert, which the profiler was counting inside `expert ffn` while it ran on the host, is **20.2 ms/token on its own** -- 34% of the bucket, ~273 MB at about 13.5 GB/s. Splitting it out leaves the routed experts at 39.7 ms against the block's 11.1, so a 3.6x gap remains unexplained and is now the next question rather than a guess. Folding the shared expert into the phase-parallel block was retried on the strength of that number and produced wrong output on mixed widths, so it is not shipped: uniform 1408 and uniform 2816 both agree with the oracle, mixed does not, and the cause is not yet found
2026-07-30	The shared expert joins `moeFfnBlock` as one more expert, with per-expert hidden widths. Expert FFN 60.0 -> 43.2 ms/token, token 121.2 -> 95.8	It is a 2816-wide dense FFN on every layer against a routed 1408, and on the host it cost 20.2 ms/token -- a third of the expert-FFN bucket at about 13.5 GB/s. Folding it lost when the block was a per-expert chain, because it moved work onto a path where every step waited behind a barrier; against the four-phase block it is simply a seventh column and wins. The mixed widths this requires were reported wrong in an earlier attempt, which is why every shape handed to a kernel is now per expert and a test covers 1408 beside 2816 against the exact oracle: uniform-1408, uniform-2816 and mixed all agree, at three different router-weight combinations. Isolating each expert with the other's weight zeroed -- both correct -- is what showed the earlier failure was a transient build rather than the design
2026-07-30	The 3.3x between the MoE block in isolation (~13 ms/token) and in the engine (43 ms/token) is *not* GPU clock ramp and *not* the working set. A 1 ms host gap between calls costs 8%; rotating a 698 MB set of experts so nothing is reused costs 3%	Both were plausible and both are now measured rather than argued. The block was exercised at the shape a layer actually issues -- seven experts, 35 MB per call -- back to back, with a host gap matching a layer's attention, and against a working set twenty times its own size. It holds 71-85 GB/s in every case. So the remaining difference is neither how often the engine calls it nor what it reads, and the next candidate is the one structural thing left untested: the engine addresses every expert as an offset into a single multi-gigabyte arena buffer where the benchmark uses one small allocation each. That probe is written and does not yet run on this machine, so it is recorded as the open question rather than as a finding

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2026-07-30	The profiler counts MoE layers that took the backend block against those that fell back to the host, and reports 26.0 against 0.0 -- the fast path is taken on every layer. A fifth explanation for the isolation/engine gap, arena buffer size, is also refuted: the same experts placed a gigabyte deep inside a 2 GB mapping run at 77.6 GB/s	The counter exists because a path declining silently has cost this project two long detours -- the compute seam that never exported `moeFfnBlock`, and the missing Q5_0 pipeline that made the block refuse every layer of the one checkpoint it was written for. Both looked like slow code and were unused code. It is worth the two lines to never ask that question by inference again. What remains unexplained is the block measuring ~12 ms/token in isolation under every condition tried -- back to back, with a host gap, against a 698 MB rotating working set, inside a 2 GB arena, all 71-85 GB/s -- and 35.6 to 68.5 ms in the engine across runs. The one structural difference left untested is that the engine's experts are file-backed pages of the store mapping where every benchmark used anonymous memory; the heap-cache configuration that would isolate it cannot be compared without a warm-up long enough to retire the verify transient, which this machine's drift then swamps
2026-07-30	MLA attention has a Metal kernel and a device-resident compressed cache (`mla_attn.metal`, one head per threadgroup on the absorption identity). It runs, it is correct, and it is **slower**: attention 42.2 ms/token on the device against 17-24 on the host, and the token 114.6 against ~85. Left opt-in under `--gpu-ops`	Built to test a specific claim -- that loom runs DeepSeek on the host because MLA attention had no kernel, so the residual stream could not stay on the device and every layer round-tripped activations across the bus. The kernel closes the first half of that and the claim is still not supported: submissions go 26.0 to 53.0 per token and the token gets slower, because each attention layer is its own command buffer ending in `commitAndWait`, which is not residency. Two silent gates had to be removed before it dispatched at all, both the same failure already fixed elsewhere in the file: allocating at model load before the Metal context exists, and then being handed the model's native 163,840 context rather than the capped 4,096, so it declined on shape. Each looked exactly like "unsupported shape". The open question is unchanged and now better posed: attention on the GPU was never the argument, keeping the token there was, and testing that needs the recorded whole-token path this kernel is a prerequisite for. Verified against an f64 oracle at 37 positions on a non-zero layer, and checked by dropping the rope term -- which only makes attention position-blind and would otherwise read as slightly worse text

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2026-07-30	Read from llama.cpp's source rather than inferred: `ggml_metal_graph_compute` encodes an entire graph into **n_cb + 1 command buffers** (n_cb is 1 at init, so about two per token) and calls `waitUntilCompleted` once per graph, splitting nodes across threads by `n_nodes_per_cb`. loom issues **53 per token** and drains the GPU on each	This is the difference, and it was available to look up for the whole of the preceding work. It also explains the result immediately before it: putting MLA attention on the device added 27 more command buffers, each with its own `commitAndWait`, so it added drains rather than residency and duly ran slower. The profiler had been reporting the mechanism as a cost to accept -- submissions 26 -> 53, ~13.9 ms -- rather than as the thing to remove. The reason llama.cpp *can* encode a whole graph is the second half: ggml computes MoE routing on the device as graph ops, where loom computes it on the host and so must synchronize every layer. The plan that follows is therefore not "port `layerBlock` to MLA": record into the existing `beginFrame`/`endFrame` seam that the MLA and MoE paths currently bypass, move routing onto the GPU so the per-layer sync goes away, and only then is a comparison with llama.cpp like-for-like
2026-07-30	Fusing the absorption into `mlaAttnHeads` -- two dispatches, one command buffer, `q_abs` never returning to the host -- takes submissions from 80 to 53 per token and 118.8 to 98.5 ms. Still above the host path's ~85, because 53 buffers is still 53 waits	The first change in this direction to move the number the right way, and it confirms the cost tracks submission count rather than kernel quality. What it also shows is that the remaining halving is not more of the same: to put the MoE block in the same buffer as the attention before it, the routing between them must leave the host, and `llama-graph.cpp`'s `build_moe_ffn` says what that actually requires. Top-k there is a device graph op (`ggml_argsort_top_k`), the experts are fetched by `ggml_mul_mat_id(ctx0, w, cur, ids)` -- one 3D tensor indexed by an id tensor on the device -- and the selected indices are never read back. So "routing on the GPU" is not a step on its own: loom's MoE block takes explicit per-expert host pointers, and it would need a `mul_mat_id` equivalent before device-side routing buys anything. The distributed path complicates that further, since its experts are arbitrary offsets into a store rather than planes of one tensor, and would need an offset table on the device or to stay host-routed
2026-07-30	Expert selection moves to the device: `moe_route` (scoring, bias, top-k, renormalization -- `ggml_argsort_top_k`'s role) and id-indexed matvec kernels for q4_k, q5_0 and q8_0 (`ggml_mul_mat_id`'s shape: one 3D tensor, planes picked by a device-side id buffer,	While the host picks the experts there is a read-back between a layer's attention and its FFN, so the two cannot share a command buffer whatever else is recorded. The earlier q5_0 failure -- slot 0 right, slot 1 on wrong -- was not the kernel: the test freed and reallocated fixtures at

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	per-slot activation stride for the down projection). Each verified bit-for-bit against the plain kernels across six out-of-order planes, and the router against `moe.route` across all eight gating/bias/normalization combinations	the same address and `wrapFor`'s cache handed the GPU a stale mapping of freed memory. A content check for that was written and reverted as vacuous (the buffer *is* the host memory through the same virtual address), so it is stated as `wrapFor`'s contract instead: wrapped memory lives as long as the context
2026-07-30	W_v joins the attention command buffer as an id-kernel dispatch with identity ids -- head h's value rows are a plane at stride (nope+vd) rows, offset nope rows in, its input its own o_latent, which is exactly the per-slot x-stride shape built for the MoE down projection. One dispatch replaces sixteen per-head host matvecs, the last host step inside attention. On the first clean machine of the effort (rebooted; swap had sat at 8-10 GB through every prior measurement): **18.2 tok/s under --gpu-ops against 8.7 on the host path, identical text** -- attention 24.7 -> 11.8 ms/token, expert FFN 68.0 -> 30.7	The first same-conditions measurement where the GPU path wins end to end, and the margin is the accumulated structure rather than any one kernel: absorb+attention+W_v as one submission, route+experts+reduce as another, everything resident. The oracle test extends through W_v -- a q4_k fixture with the real plane layout, f64 reference -- so the whole attention buffer is covered by one assertion. Against the 38 tok/s bar (llama.cpp on an M3): 18.2 on an M5 with ~53 submissions per token still standing, and the remaining path -- fusing the two per-layer buffers, then the cross-layer frame -- is the measured-not-argued route to the rest
2026-07-30	`mlaLayerTail`: a whole MLA layer -- absorb, attention, W_v, output projection, residual, FFN norm, router, routed MoE, second residual -- as **one submission**. 54 -> 28 command buffers per token, and warm decode reaches **35.5 ms/token (~28 tok/s)** against 54-64 without the tail, identical text. Pinned by a differential test against its own constituent buffers plus host glue	Three more silent gates had to fall before the tail fired at all, the fourth through sixth of the series: `gguf run` never called `mlaInit` (fixed with the capped context, since the native 163,840 declines on shape), never called `uploadAbsorbWeights` (so `li >= mla_wk.items.len` declined every call), and the router is one of the checkpoint's f32 tensors, for which no dmmv kernel existed -- a 15-line `dmmv_f32.metal` closes it. Each presented identically: correct text, no error, submission counter unmoved -- which is why `cb/tok` is now printed in the compact profile line and `LOOM_NO_TAIL` exists to bisect the tail against the two-buffer path inside one binary. Against the 38 tok/s bar: ~28 warm, with the cross-layer frame (28 -> ~2) the remaining structural step
2026-07-30	`mlaTokenFrame`: the whole token -- every layer's attention head (norm, q/kv_a projections, cache norm, rope), attention, and FFN, then the final norm and lm_head -- as **one command buffer**. 28 -> 1.8 buffers per token, **27.9-30.4 ms/token against the tail's 38.7-47.6, identical text: ~34.7 tok/s wall-clock including prefill**, against the 38 tok/s bar from llama.cpp on an M3	This is `ggml_metal_graph_compute`'s shape reached: the residual stream lives in a device slot for the entire token and only logits return. The head coming off the host needed the previously merged prerequisites -- the rope kernel oracle-tested against `ropeApply` itself, strided q in absorb/attention, `copy_f32` -- plus the frame writing the compressed cache on-device with the rows mirrored back after each token, so a later fallback can never attend over stale positions. One trap nearly repeated the day's oldest mistake in reverse: the first frame mea-

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		surement said *slower* (83.6 ms) and was the cold run paying pipeline compiles and every tensor's wrap; warm and alternating, the frame wins by ~30%. The frame declines -- and the engine falls back per layer -- on anything it cannot express: a router bias, the q-LoRA path, a type without a kernel
2026-07-31	A frame phase-split (`LOOM_FRAME_DEBUG`: each layer as four waited sub-buffers, per-category sums) found the absorb kernel launching **sixteen threadgroups** -- one per head -- on a device that wants hundreds, every thread crawling tens of kilobytes serially. Re-gridded to one SIMD group per output element (8,192 groups on the real model): **41.4-42.9 tok/s, past the 38 tok/s llama.cpp-on-M3 bar**, identical text	The rewrite briefly reproduced the codebase's canonical bug in a third place: three call sites dispatch the absorb pipeline, the two single-line ones were updated mechanically, and the multi-line one in `mlaAttnHeads` kept the old grid -- with a threadgroup of 64, the new kernel computed a fraction of the outputs and left the rest stale, which is a plausible residual stream. Both the f64 oracle and the layer-tail differential failed on it immediately, which is the whole reason they exist. The phase ledger that motivated the change: head 10.0 ms, attn 13.7, proj 9.2, ffn 33.1, lm_head 2.9 in split mode, against ~1 ms of actual matvec work in the head -- occupancy, not bandwidth
2026-07-31	Three squeezes from the same ledger: the attention weighted-sum re-gridded to one SIMD group per output element (the fused kernel kept sixteen threadgroups for the bandwidth half of attention -- fine at seq 40, collapsing at seq 4,096); W_k stored f16, halving the largest reducible read outside the experts (~113 -> ~56 MB/token); the six per-slot SwiGLU dispatches merged into one and a provably-false barrier removed. Best run **44.4 tok/s**, band 39.7-44.4 against the prior 41.4-42.9	At a 40-token context these are ~1 ms effects and the bands overlap; the wsum re-grid is the one that changes the long-context story, since the O(seq*kvr) sum is what grows. The attention split reproduced the canonical bug an eighth time -- the layer tail's own attention dispatch, a third call site again, kept the fused semantics and fed probabilities to W_v as if they were o_latent -- and the layer-tail differential caught it before any manual run, same as the seventh. f16 rounding of W_k passes the f64 oracle unchanged: rounding a weight already quantized to ~4.5 bits is noise against the quantization itself
2026-07-31	The Vulkan backend exists (issue #13): a dlopen-loaded device layer (~25 functions, hand-declared -- no SDK at build, no driver required at run, declines cleanly without either), a GLSL Q4_K matvec compiled to committed SPIR-V, and a seam-complete backend that resolves everything to the CPU except that one kernel. **All 83 tests pass on llvmpipe** on the Hetzner box, including the f64 oracle with silent-fallback detection	Correctness only, by explicit agreement: llvmpipe is software Vulkan and the only Vulkan this project currently has, so every performance decision waits for physical hardware. The bring-up deliberately repeats the Metal backend's day one -- hottest kernel first, oracle before use, everything else declining to the CPU -- because the expensive lessons (silent gates, fallback detection, pinned scales, alignment) were already paid for there; the SPIR-V alignment panic on the first run was caught by the test in one round for exactly that reason. dlopen rather than -lvulkan is what lets the ma-

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		cOS host cross-compile the Linux test binary it cannot itself run
2026-07-31	Vulkan grows to the full dmmv family -- q5_0, q8_0, q6_k and f32 GLSL kernels alongside q4_k, ported from the token-exact CPU/Metal arithmetic -- and weights become device-persistent: a pointer-keyed upload-once cache (length mismatch re-uploads; mmap'd tensors make staleness impossible) replaces the bring-up's re-upload-per-call. The oracle test now loops all five types and uses the submission counter, not output comparison, as fallback detection	Pre-rental threshold work: these are the kernels DeepSeek-V2-Lite's token path actually touches (q4_k/q5_0/q8_0 experts and projections, q6_k lm_head, f32 router), so a green fused-layer differential on llvmpipe -- not a rented GPU -- is the gate for renting one. Memory-type choice stays behind `Device.alloc`, the single seam where DEVICE_LOCAL + staging lands when real hardware exists to measure it
2026-07-31	The Vulkan routed-MoE chain is complete and differentially verified: `moe_route` (score, bias-shifted top-k, unbiased gates, f16-clamped renorm), id-indexed matvecs for q4_k/q5_0/q8_0 (`ggml_mul_mat_id`'s role), device SwiGLU and gated reduce, shared expert -- ids and gates never return to the host between route and reduce. Verified against the shipped host router and an f64 dequantize-everything reference exercising all three id kernels plus the shared expert; **85/85 on llvmpipe**. This meets the pre-rental threshold	One bring-up difference from Metal, deliberate: each step is its own submit-and-wait rather than one recorded command buffer, because the minimal device layer only has `dispatchWait`. The kernels are what a real GPU will run; the submission shape is not, and fusing submissions is a performance decision that waits for hardware. With the differential green, renting an NVIDIA box is now justified: first hardware task is DEVICE_LOCAL memory + staging behind `Device.alloc`, then measuring
2026-07-31	**MLA attention on Vulkan**: the compressed-cache attention (absorb, scores+softmax, weighted sum, W_v as an id dispatch with identity ids) and `mlaLayerTail` -- attention through output projection, residual, FFN norm, router and the routed MoE chain as **one recorded submission** per layer. New GLSL: mla_absorb, mla_attn, mla_wsum, rmsnorm, add_vec; id kernels grew a `base` push constant (Vulkan binds whole buffers, so the W_v row offset and per-layer cache offsets ride in push constants; the c/k_rope caches share one buffer for the same reason). Differentials: attention vs an f64 dequantize-everything reference with decoy rows in the other layer's cache region; tail vs its verified constituents. **87/87 on both the RTX 3060 and llvmpipe.** End to end on the 3060: 7.1 tok/s over 128 tokens (2.3x CPU), ~115 ms/token marginal decode, cb/tok 59	The submission ledger, as always: the tail collapsed a layer's ~15 waits into one, and the decode profile now splits ~75% attention-bucket (26 tail submissions x ~2 ms each of submit-and-wait overhead) -- on this discrete GPU a submission costs ~2 ms against Metal's ~260 us, which makes cb/tok 59 the whole story. The two remaining ports are known quantities: the whole-token frame (Metal: cb/tok 1.8) to collapse 59 submissions to ~3, and GPU prefill (the prompt still runs the CPU batched path at ~690 ms/token, which is why short generations read slower than the marginal rate)
2026-07-31	**The whole token as one Vulkan submission** (`mlaTokenFrame`, the port of Metal's): per layer the attention head (attn norm, q/kv_a projections, kv_a norm straight into the cache row, rope on q and on the cached k_rope -- new	Two lessons paid for. First: an intermittent 1-in-5 wrong-output on the 3060 was misread as a GPU barrier race for a day -- widening the barrier to ALL_COMMANDS "fixed" a 10-run sample, and a 30-run soak then showed 6

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	`mla_rope` and `copy_f32` kernels, ropeApply-faithful with YaRN), the compressed-cache attention, projection/residual, and the FFN (routed-with-shared-expert or dense), then final norm and q6_k lm_head. **cb/tok 59 -> 1.0**; 3060 end-to-end 7.1 -> **9.7 tok/s** (tail path same binary via LOOM_NO_FRAME: 6.0), ~90 ms/token marginal at 90%+ GPU utilization, weights fully VRAM-resident (10.1 GB). Verified by a whole-token f64 differential (two layers, dense + MoE-with-shared-expert, cache rows checked because a frame that attends right but caches garbage only fails on the *next* token) plus a determinism probe; 89/89 on the 3060 (30-run soak, 0 failures) and llvmpipe (15 runs)	failures anyway. The real bug was the pointer-keyed weight cache serving stale device tensors when TESTS free and reallocate same-shape buffers at recycled addresses; mmap'd model weights cannot hit it, tests now clear the cache, the barrier went back to its spec-correct compute scope, and the soak is the only sample size that counts. Second: with one submission per token the profile stops being about submission overhead at all -- 90%+ utilization at ~90 ms/token means the correctness-shaped kernels themselves (scalar loops, one workgroup per row) are now the frontier, along with CPU prefill. That is where Metal's occupancy lessons finally apply to Vulkan, with hardware to measure them on
2026-07-31	**Vulkan kernel-efficiency pass**, driven by a ported LOOM_FRAME_DEBUG phase split (head/attn/proj/ffn/lmhead sums per token). Round one: the whole dmmv family vectorized -- weights as u32 word loads (an unaligned-tail-safe helper for the 22/34/210-byte block formats, buffers padded a word), activations as vec4 -- 9.7 -> 15.6 tok/s, ffn phase 54 -> 23 ms, lm_head 5.5 -> 2.0. Round two: absorb and wsum re-gridded from one-workgroup-per-output-element (every read a 64-way scatter) to one workgroup per (head, 64-output tile) with threads walking rows serially -- the coalesced shape -- plus vec4 attention dots and a 256-thread rmsnorm: attn phase 14.1 -> 5.4 ms. **End to end: 19.8 tok/s over 256 tokens; marginal decode 36.6 ms/token = 27 tok/s.** 89/89 on the 3060 and llvmpipe	The two rounds are the two classic GPU sins, caught in order of cost: uncoalesced/narrow loads (byte-at-a-time from a 360 GB/s device) and scatter-shaped grids (the same re-grid Metal paid for at 16-threadgroups-per-absorb). The oracle suite is what made both rewrites safe to do in an afternoon: every kernel's arithmetic is pinned by an f64 differential, so a vectorization slip fails a test instead of shipping plausible text. Remaining, in measured order: the MoE id kernels still run ~57 GB/s effective (row-batching, NR0=4 as Metal does, is the known next step), and prefill still runs the CPU batched path
2026-08-02	**Three pre-configured networks: devnet / testnet / mainnet** (ids 1337 / 2 / 1 -- protocol constants now): `--network devnet` resolves the stable id and the canonical model from an in-binary registry, Ethereum's chain-registry shape for LLM networks. devnet = GLM-4.5-Air (106B-A12B, ~66 GB Q4 -- verified "loads and runs", distributable across the existing three-machine fleet with zero new hardware); testnet = GLM-4.6 (357B-A32B, ~200 GB Q4, verified likewise on the existing glm4moe engine); mainnet = GLM 5.2 (glm-dsa, gated on the engine variant), with Kimi K3 the recorded alternative. Arch policy enforced at startup: testnet/mainnet REFUSE a mismatched model, devnet warns	One-network-one-model graduates from convention to configuration: the registry makes the invariant that RAG text-only gossip and expert fetch both lean on into something a node enforces before serving a single token. The phase ladder is deliberate -- every claim validates on devnet at 106B on hardware already paid for, performance proves out on testnet at 357B on one 256 GB box, and mainnet waits for exactly two named gates (the glm-dsa engine, task #28, and a hardware approval) rather than an open-ended "someday"

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2026-08-02	**GLM 5.2 readiness audit** (docs/GLM52-READINESS.md, run with loom's own `gguf check` against live checkpoints): GO with one bounded engine variant. GLM 5.2 GGUFs exist (2-bit ~239 GB to 4-bit ~475 GB); the arch is `glm-dsa`, expert-aligned -- loom's sharder distributes it already; llama.cpp mainline itself runs it with a dense-attention fallback, so loom's delta is a deepseek2-family variant (MLA, noaux_tc, shared expert all in hand) that maps the tensors and skips the DSA indexer + MTP at load. And the bridge is free: **DeepSeek-V3-0324 Q4_K_M (671B, 256 experts) passes `loom gguf check` with "loads and runs" on the existing deepseek2 engine** -- thesis-scale distribution with zero engine work, awaiting only a hardware decision (512 GB box for 4-bit, 256 GB for 2-bit)	The audit method is the point as much as the result: fifteen minutes with `gguf check --header-only` against remote URLs answered what could have been a week of speculation -- the tool built for integrity checking doubles as an architecture-compatibility probe. DSA sparse attention is deliberately deferred as a speed feature, exactly as llama.cpp mainline treats it
2026-08-02	**The three-machine test** (Hetzner CPU origin 100%, Canadian RTX 3060 node 33%, NAT'd macOS laptop 20%, v0.12.0 binaries): expert-sharded serving verified against the manifest root on all three; wrong-network node (--network-id 999) dialed the origin for 45 s and never entered the mesh; RAG chunks propagated in every direction including the NAT pull; and the full loop closed -- facts ingested only on the laptop were gossiped out, retrieved by the origin (prompt 28 -> 81 tokens with prepended context) and answered correctly by the distributed model, where the same question minutes earlier had produced a confident hallucination. Churn recovery observed live: the origin was OOM-killed mid-test and both joiners redialed and re-formed without intervention	Four findings, each now owned: (1) the network-id auto-derivation ran before the store existed, so every node silently derived id 0 -- consistent, so the mesh still gated correctly, but the manifest default was dead; fixed, computed after the store is final. (2) The RAG store is memory-only and the OOM restart wiped every node's chunks -- persistence filed. (3) The origin footprint at --ram-gb 3 re-confirmed the documented 0.5 guidance for small boxes. (4) Cross-continent expert fetch inside the token loop is as slow as the design doc's fabric table predicts -- WAN distribution buys capacity, not speed; the RAG-augmented longer prompts amplify prefill fetch cost, which argues for the shared-prefix-KV feature already sketched
2026-08-02	**RAG pull direction (RagInvReq, 0x23)**: planning the three-machine test surfaced a protocol hole -- the exchange was push-on-dial only, so a NAT'd node (a laptop) could send chunks on its outbound dials but never receive any. One dial now converges both directions: push (Inv -> Want -> Push) then pull (InvReq -> Inv -> Want -> Push), all server handlers stateless	Found by walking the deployment before running it -- the Mac in the three-machine topology can only dial out, and the propagation asymmetry would have shown up as "gossip works but my laptop never gets chunks". SPEC updated
2026-08-02	**The vector index extracted to zigstack/vector-index** and consumed as a package: the flat exact scan (implementation of record, static everywhere) and the FAISS dlopen accelerator now live in their own repo with their own CI and an agreement test between the two paths;	Shape 3 of the make-FAISS-a-dependency discussion, chosen over vendoring FAISS's C++/BLAS/OpenMP stack: the package is where HNSW lands when ANN scale demands it, benefiting loom and any other Zig consumer,

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	loom's store keeps text, hashes, compression and gossip, and delegates vectors and search to the dependency. Vector storage improved in the move -- one contiguous row-major buffer instead of a slice per chunk, the scan's cache locality	and the accelerator/reference agreement contract travels with the code that must honor it
2026-08-02	**Brotli vendored as a build-time dependency** -- the pinned v1.1.0 C sources compile into every release target through build.zig (pure C, zero dependencies, MIT), so at-rest chunk compression is always available and the dlopen probe is gone; the bindings became two linked externs and the test now asserts compression actually happens, not merely that the path degrades gracefully. FAISS stays dlopen by explicit contrast, recorded in build.zig itself: C++ + BLAS + OpenMP has no place in a four-target musl-static cross build, and at loom's 64K-chunk cap the exact scan is the same algorithm FAISS-Flat runs -- FAISS engages when installed as a BLAS-accelerated flat search, and its ANN families (the reason to ever vendor more) would be better answered by an HNSW in Zig if scale demands it	The rule this pair sets for future native code: vendor what Zig's own toolchain can cross-compile statically everywhere; dlopen what cannot, with a pure-Zig fallback that keeps the feature testable on CI. Verified: musl x86-64 and arm64 cross builds and the native suite all green with brotli linked
2026-08-02	**RAG storage and convergence, answered precisely**: (1) FAISS stores nothing compressed here -- the flat inner-product index holds raw f32 vectors by design (its PQ/SQ compressed families are unwired), and chunk *text* never enters FAISS at all. (2) Compression now exists at both layers where it pays: the wire already ran every RAG frame through the frame encoder's adaptive snappy (kept only when smaller); at-rest chunk text is now brotli-compressed when `libbrotlienc/dec` are present -- dlopen, the FAISS/Vulkan loader trade -- with raw storage as the fallback. At-rest format is purely local: hashes are of raw text and Push carries raw text, so mixed-library networks interoperate unchanged. (3) Chunk additions ARE a global topic: the Inv/Want/Push exchange runs against every known peer each gossip round plus transitive re-gossip -- and the inventory window now ROTATES when a store exceeds the 512-hash round cap, closing the hole where a rejoining peer could never learn chunks older than the newest window. Every hash is advertised within ceil(count/cap) rounds; a test pins the full-coverage property	Brotli over snappy for at-rest text is the right split: snappy stays the wire codec (speed, already adaptive), brotli's ratio wins on stored prose, and neither becomes a dependency -- both decline to raw cleanly. The rotation bug is worth its whitepaper line: "gossip to all peers" was true per-round from day one, but eventual convergence for late joiners needed the window to move

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2026-08-02	**Gossiped RAG chunks with FAISS-accelerated retrieval** (`--rag`): nodes keep a chunk store searched by cosine before every prompt reaches the engine (top-k prepended as context); new chunks -- ingested via `POST /v1/rag/chunks` or learned from peers -- reach the whole network through an Inv/Want/Push exchange piggybacked on the gossip round. FAISS rides behind dlopen (`libfaiss_c`, the Vulkan-loader trade: no build dep, no runtime requirement) with an exact-scan fallback that is the reference implementation and the CI-tested path. The load-bearing design choice: **only text travels**. One network serves one model (network_id), so every node recomputes the identical embedding -- mean-pooled token_embd rows, L2-normalized -- from the same text; vectors are never accepted from the wire	The two features lock together deliberately: network_id is what makes text-only RAG gossip sound, because embedding equality across nodes is exactly the same-model guarantee the membership gate enforces. Poisoning reduces to what it should be -- a peer can contribute bad *text*, visible and dedupable, never a bad vector behind good text. Caps bound every round (512-hash inventory, 32-chunk push, 8 KB chunks, 64 K store); eventual convergence across rounds. Embedder quality is the known v1 trade: mean-pooled input embeddings are crude, and swapping in a real embedding model later changes only `setEmbedder`
2026-08-02	**network_id: chainId semantics for LLM networks** (wire proto v2): one loom network serves one model, and every Heartbeat/Announce now carries a u64 network identity. Peers on a different network are refused (`ERR wrong_network`) at the p2p surface and dropped at gossip merge, so a record from another LLM network can never enter the mesh table, however it arrived. `--network-id N` configures it; the default derives from the weight manifest's leading eight bytes, so nodes sharding the same model agree with zero configuration	The split of duties is deliberate and mirrors Ethereum: network_id gates *membership* (stable across hardforks when set explicitly), manifest_version keeps gating *content* (expert fetch already refused cross-version requests). The planned ENR record gains network_id alongside the manifest fields. SPEC.md updated as the governing document
2026-08-01	**The network registry carries default bootnodes, and joining the devnet is one command**: each named network entry may list bootnodes; `--network <name>` with no explicit `--bootstrap` and no local GGUF dials the registry default (printed as such), so `scripts/join-devnet.sh` reduces to install-if-missing plus `loom node --network devnet` with NAT-safe defaults (hold-fraction 0.2, RAG on, advertise unset). Explicit `--bootstrap` and origin mode (`--gguf`) always win over the registry default	A network whose id, model, and entry point all live in the binary makes "join the devnet" a zero-configuration act, which is what a PoC network needs to grow; keeping the override order (explicit flag > registry) preserves every existing workflow. Devnet's registry bootnode is the Hetzner box (23.88.108.96:8771)
2026-08-01	**Windows becomes a release target** (`x86_64-windows-gnu` cross-build, published as a zip and smoke-gated on a real Windows runner before publish; Docker images stay linux/amd64 + linux/arm64): five portability seams closed -- argv arrives as WTF-16 and is decoded	The port touched only OS seams the code had already isolated -- no engine, wire, or storage logic changed, which is the design working as intended. The pread fallback that covers an unmappable store now also covers nothing: Windows maps too. WSL2 remains the docu-

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	through the std `Args` iterator; RSS reads peak working set; the store's read-only weight map uses `CreateFileMappingW`/`MapViewOfFile`; monotonic timing goes through a shared `nowMonoNs` (`QueryPerformanceCounter` on Windows, `clock_gettime` elsewhere); and both dlopen probes grow `LoadLibrary` branches (`vulkan-1.dll` here, `faiss_c.dll` in vector-index v0.1.1)	mented route for anyone wanting the Linux build
2026-08-01	**SwiGLU folded into the expert down projection** (pipeline layout widened to five bindings): the fused q5_0/q8_0 down kernels compute silu(gate)*up where it is consumed -- one dispatch and one full drain fewer per MoE layer. Best 192-token average yet (39.5 ms cumulative); 89/89 on both platforms over three runs. **This closes the Vulkan single-node optimization phase.** Final state: ~32 tok/s end-to-end, marginal in the 13-15 ms band (~70 tok/s), past the M5 Metal path's 44.4, at ~60-65% of llama.cpp's 112 on the same card	The phase ledger, complete: 0.6 -> ~32 tok/s across 17 merged PRs; every kernel f64-oracle-pinned; ten negative results recorded; the remaining ~1.5x localized to ~300 barrier drains whose no-barrier ceiling (8.5 ms = 117 tok/s) is proven on this card. The recorded path there -- norm-into-consumer fusion, then the layer-megakernel shape (llama.cpp's ~7-node granularity) -- is an architecture project, deliberately deferred: the local path now exceeds the roadmap's needs, and v1's distributed work (the thesis) resumes with this foundation
2026-08-01	**The value side fused: wsum + W_v in one kernel per head** -- o_latent accumulates in shared memory (its only consumer is the same workgroup) and the W_v rows read the quantized kv_b plane directly, q4_k and q8_0 variants from one template so the existing differentials exercise the shape. One dispatch, one buffer and one full drain fewer per layer; with the absorb fusion, the attention block is down from four dispatches/four barriers to two/two. All 89 tests on both platforms, generation verified	Where the fusion series stands against the 8.5 ms / 117 tok/s no-barrier ceiling: attention is consolidated; the remaining barrier mass sits in the FFN chain (route/gate-up/swiglu/down/reduce, structurally serial) and the head (norms, rope, projections). The norm-into-consumer fusion and a 5-binding pipeline layout (to fold swiglu into the down kernel) are the next two items; past those, the ceiling likely needs the layer-megakernel shape rather than pairwise fusion. Marginal sits in the 13.8-15.7 ms band run-to-run on this box -- thermal variance now exceeds single-fusion deltas, itself a sign the easy wins are spent
2026-08-01	**Absorb fused into the attention kernel** (dispatch-consolidation series, opener): q_abs computes into shared memory inside the per-head attention workgroup -- its only consumer -- deleting the separate absorb dispatch (27/token). Net time neutral: the deleted barrier was re-spent explicitly ordering the cache-row rope before attention (a 3060-only race the frame differential caught deterministically -- llvmpipe green, NVIDIA red, the cache-row check firing exactly as designed). Five suite runs clean after the fix	Bookkeeping for the series: each fusion must remove a BARRIER, not just a dispatch, to collect the ~15 us; the next pairs (wsum+W_v, norms into consumers, rope into the head group) are chosen on that criterion. The 8.5 ms no-barrier ceiling (117 tok/s) stands as the target
2026-08-01	**Integer-quantized activations: complete, correct, neutral -- and the measurement**	The remaining 1.6x is now ONE quantified thing: ~350 full-pipeline barrier drains per to-

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	that names the endgame: quant_act (f32 -> q8_1-style blocks) + an integer-dot q4_k id kernel where the masked quant word IS the packed i8x4 (dotPacked4x8, zero decode ALU), capability-probed (VK_KHR_shader_integer_dot_product; llvmpipe untouched), env-gated LOOM_VK_INT8 for gate/up. Generation fluent with the expected int8-class divergence; performance neutral -- those kernels were already bandwidth-saturated, confirming Ampere's spare ALU was never the constraint. The decisive by-product: re-running LOOM_NO_BARRIER at today's speed shows **8.5 ms/token without barriers = 117 tok/s -- llama.cpp's exact territory**. Barriers now cost ~5.4 ms/token (39%), up from 7% relatively when tokens took 35 ms	ken at ~15 us each. The fix is dispatch-count consolidation -- fusing absorb into attention, norms into consumers, fewer/wider ops per layer, llama.cpp's ~7 nodes per layer against loom's ~20 dispatches. That is the whole remaining roadmap to 112
2026-08-01	**Reduce fused into the residual stream**: `moe_reduce_dev` gains an accumulate flag; the frame's routed reduce writes straight into x, removing one dispatch and two barriers per MoE layer (the standalone routed block keeps fresh-write semantics, flag 0). Verified over three suite runs on the 3060 plus llvmpipe; marginal holds ~14 ms. First of the small-dispatch-tail series	The tail-shrinking continues (adds and copies remain), and the integer-activation kernel family (llama.cpp's q8_1 activations + packed integer dots) is the other named remainder on the way to 112
2026-08-01	**q5_0/q8_0 padded quants-first at upload** (22 -> 24 and 34 -> 36 byte strides, quants first so every load is a direct word read; a `paddedLen` helper corrects every plane-stride and row-bytes computation, which the q6_k round showed must be type-keyed at every site). Kernel ticks: q5_0_id −15%. The first measurement looked like a 10 tok/s regression -- the repack loop (millions of 16-byte memcpys through fresh page allocations) had doubled warmup; rewritten with fixed-size copies into a reused scratch buffer. **Marginal decode ~14 ms/token (~72 tok/s), best yet**; end-to-end 31.7 with warmup recovered	The gap to llama.cpp's 112 now decomposes to the two named remainders: their integer-quantized activations (q8_1 activations against quantized weights with packed integer dots -- loom dots f32 activations throughout) and the long tail of tiny serialized dispatches between full barriers. Both are on the queue in that order
2026-08-01	**q6_k blocks padded to word alignment at upload** (next item on the profile's ranked list): the 210-byte on-disk block defeats aligned loads, so the device copy pads each block to a 224-byte stride (+7% VRAM on the affected tensors) and the kernel's quant loads become direct word reads. Kernel ticks -16%; end-to-end holds the 32-34 tok/s band. The build also caught a classic: the first padded version	Two lessons: a padding scheme must be keyed by tensor *type* at every site, not by call path -- one helper, no exceptions; and an oracle can be green while the model is broken if the bug lives in which-buffer-was-uploaded rather than in kernel arithmetic. The ranked list continues: q5_0/q8_0 id down kernels (same padding treatment applies -- 22 and 34-byte blocks), then the serialized chain

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	repacked only the matvec/lm_head sites while Q4_K_M stores several *attention* tensors as q6_K -- the oracle passed (its path padded correctly) and real generation printed garbage. Every WeightRef now routes through one type-aware fetch, and the generation check is part of the ship gate alongside the suite	
2026-08-01	**Post-VRAM profile acted on**: with the working set device-local the kernel table transformed -- the f16 gate/up kernels now run ~250 GB/s -- and exposed `moe_route` costing 0.5 ms/token: 26 single-thread sigmoid loops per token. Scoring now parallelizes across the workgroup (selection stays serial, as designed). **34.4 tok/s end-to-end, ~14.5 ms marginal (~69 tok/s)**; 89/89 both platforms	Ranked remainder by the same table: q6_k (16%, 210-byte blocks defeat uvec4 alignment -- needs its own shape), the q5_0/q8_0 id down kernels (15%, still word-assembly loads), then the frame's serialized chain. Each now has a per-kernel tick count to be judged against
2026-08-01	**The working set moves to VRAM -- the answer the whole ledger was pointing at**: only the *weights* were device-local; every activation, expert slot, attention intermediate, and the 226 MB compressed cache lived in host-visible system RAM, so all ~600 dispatches per token paid PCIe first-touch latency on their inputs -- a tax no kernel shape could hide, which is why ten shape experiments measured neutral. Phase 1 (intermediates device-local, `download` primitive for the non-frame readbacks): 20.6 -> 27.6 tok/s, id kernels collapse from 26% to 2% of GPU time. Phase 2 (cache device-local, a host-visible mirror strip fed by two in-frame copies so the engine's post-frame readback stays a memcpy): **32.1 tok/s end-to-end, 15.7 ms/token marginal = 63.7 tok/s** -- decisively past the M5 Metal path's 44.4. 89/89 on both platforms, three suite runs on the race-sensitive changes	The instrument that found it was the per-kernel profiler showing a *uniform* deficit across all kernels -- the signature of an environmental cost, not a kernel cost. The M5 comparison also closes with a symmetry: Metal never had this bug class because unified memory has no wrong side of the bus. Remaining distance to llama.cpp's 112 is now 1.8x, with the frame's serialized-chain structure the leading suspect and the same instruments in place
2026-08-01	**Metal's overlap schedule ported to the frame**: the cache-row rope now runs concurrently with the absorb, and the routed reduce with the shared expert's gate/up -- the disjoint-buffer overlaps Metal's encoder proved -- two full drains fewer per layer, verified over four suite runs (overlap bugs are race-shaped). Performance: neutral, closing the incremental ledger at ten measured experiments	The conclusion the ledger forces: within a single serialized command stream on this GPU, no kernel-granularity or barrier-granularity change moves the marginal token time. loom Vulkan stands at 21 tok/s end-to-end / ~31 marginal (6.8x CPU); the 3.6x to llama.cpp's 112 requires the deep restructure -- an execution model that keeps many operations resident (their graph scheduler's property), which is an architecture project, not an optimization pass. Until then, the M5-beating path on commodity NVIDIA runs through bigger cards: the bandwidth-proportional share of a 3090 puts current

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		loom kernels near the M3 bar without further kernel work
2026-08-01	**The llama.cpp thread map itself, ported and measured**: q4_k kernels rebuilt in mul_mat_vec_q4_k's exact shape -- 128-thread workgroups, 16 threads per super-block (v_im/v_in), contiguous warp quant loads, four rows sharing every activation read. A word-addressing bug on the first attempt produced garbage text and 3 oracle failures; fixed, 89/89. Performance: **neutral** (~49 ms avg, ~20.5 end-to-end)	The dispatch-shape hypothesis joins the eliminated list. Every level llama.cpp's kernels differ at -- arithmetic, loads, reduction, precision, thread map -- is now ported and measured neutral inside loom's execution model, which localizes their 3.6x to the execution model itself: graph-level scheduling that keeps many operations resident where loom's strictly-barriered single chain runs each small dispatch alone. Closing it means restructuring how the frame issues work (events/split-barriers, independent-op overlap), not another kernel variant. The negative-result ledger, the profiler, and the 112 tok/s reference build are the complete inheritance
2026-08-01	**Range-gated f16, done right -- and the final elimination**: a f16-product q4_k id variant used ONLY where the activation is provably post-rmsnorm (expert gate/up; products ~3e2, block sums ~1e4, under f16's 65504 -- the down projection's SwiGLU-scaled inputs stay f32). All 89 differentials pass. Performance: **neutral** -- Ampere executes fp16 shader ALU at 1:1 with fp32; the f16 win on NVIDIA lives in tensor cores, not compute shaders	This closes the shader-micro-technique ledger: NR4 +4%, 128-bit loads +6%, unpack8 0, subgroupAdd 0, f16 0 (safe form) / NaN (unsafe form). llama.cpp's 3.6x on this card therefore comes from its dispatch/tiling architecture -- spec-tuned wide workgroups covering rows with high per-thread ILP, and overlapped execution without full-pipeline barriers -- not from any per-instruction trick. That architectural rework (or cooperative-matrix use) is the remaining path to the measured 112, and it is a redesign, not a patch
2026-08-01	**f16 kernel arithmetic: attempted, rejected by the oracle**: the q4_k pair's products moved to float16_t (llama.cpp's fp16 mul_mat_vec variant, shaderFloat16 enabled, f32 per-block accumulation to bound error). The f64 differential answered with NaN -- f16's 65504 ceiling overflows on legal inputs the oracle generates, and would overflow on any real activation spike. Reverted; perf was ~neutral in the run that completed anyway	llama.cpp ships this variant accepting the overflow risk on devices it profiles as safe; loom's verification bar does not accept a kernel that can silently produce inf on legal input. Closing the remaining 3.6x to llama.cpp's measured 112 tok/s therefore requires either per-tensor range analysis to gate f16 safely, or the non-numeric routes (spec-constant shapes, finer synchronization) -- a design decision recorded here rather than made silently
2026-08-01	**The ceiling, measured**: llama.cpp's own Vulkan backend, built on the same RTX 3060 against the same DeepSeek-Coder-V2-Lite Q4_K_M, decodes at **112 +/- 14 tok/s** (llama-bench tg64). loom sits at ~31 marginal	This converts the open question -- can Vulkan on this card beat the M5's 44? -- into a proven yes with 2.5x headroom, and turns further optimization from hypothesis into diffing against a working reference on identical hardware. The major unported technique standing: f16 kernel arithmetic (FLOAT_TYPE = float16_t via shaderFloat16 -- all loom kernels compute f32), then spec-constant workgroup tuning and finer-grained synchronization. Setup note for

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		reproduction: Ubuntu 22.04 lacks glslc; the LunarG jammy repo provides shaderc, vulkan-headers and spirv-headers as separate packages
2026-07-31	**subgroupAdd reductions across the dmmv family** (the second llama.cpp structural item): the shared-memory tree -- six barriers per row-group -- became one subgroupAdd, one subgroupElect write, one barrier, one cross-subgroup pass, sized for llvmpipe's 8-wide subgroups. Measured on the 3060: **neutral** (20.3 tok/s, ~31 ms marginal), 89/89 green on both platforms	Every llama.cpp mat-vec technique is now ported and individually measured on this hardware: unpack8 neutral, subgroupAdd neutral, load width +6%, NR4 +4%. The uniform residual across all quant types survives arithmetic, reduction, and load-shape changes -- what has NOT been tried is llama.cpp's per-device spec-constant tuning of workgroup size and rows-per-workgroup, which needs vkSpecializationInfo plumbing in the pipeline layer, and their higher-occupancy shapes for small-column tensors. The scoreboard and the elimination log are the inheritance for that session
2026-07-31	**llama.cpp Vulkan analysis applied**: read ggml-vulkan's mul_mat_vec_q4_k.comp + mul_mat_vec_base.glsl at source. Three techniques identified: `unpack8` one-instruction byte unpacking, `subgroupAdd` reductions (no shared-memory tree, no barriers), and spec-constant-tuned workgroup size/NUM_ROWS. Ported the first (q4_k pair, instance bumped to Vulkan 1.2 for SPIR-V 1.5): measured **neutral** -- the driver was already fusing the shift chains, eliminating dequant ALU from the suspect list. 89/89 green	What remains from the llama.cpp playbook, now the top of the queue: subgroupAdd reductions and spec-constant workgroup shapes -- the structural differences, not the arithmetic ones. The per-kernel scoreboard (LOOM_VK_KERNEL_PROF) is the judge
2026-07-31	**A Vulkan-native kernel profiler, and what it ruled out**: `LOOM_VK_KERNEL_PROF` writes a timestamp query after every recorded dispatch (the stream is fully barriered, so consecutive timestamps are per-dispatch GPU durations) and prints per-pipeline totals -- the profiler Nsight Compute cannot be, since it profiles CUDA only. First table: q4_k + q4_k_id = 47% of GPU time at 54-75 GB/s effective, uniform across every quant type; GPU-busy ~28 ms of the ~35 ms marginal. Acting on it, the q4_k pair moved to 128-bit uvec4 loads (the 16-byte block header is exactly one uvec4) -- and gained only ~6%: 20.3 -> 21.1 tok/s end-to-end, 31.8 ms marginal. Also corrected: prefill was *already* on the GPU (the deepseek engine has no stepBatch, so prompt tokens run the whole-token frame); the startup cost is the one-time 10 GB VRAM upload, not CPU prefill	Ruled out by direct measurement, in one line each: submissions (cb/tok 1.0), residency (10.1 GB VRAM), barriers (~7% by LOOM_NO_BARRIER), coalescing (tile re-grids), x-traffic (NR4), and now load width (uvec4). The uniform ~5x-off-bandwidth across all dmmv kernels with all of those eliminated points at the dequant ALU chain and workgroup shape (64 threads, deep serial per-thread loops) -- the next experiments are wider workgroups and subgroup reductions, and they now have a per-kernel scoreboard to be judged against
2026-07-31	**NR4 dmmv + honest accounting**: the whole dmmv family (7 quantized kernels + f32)	The FFN kernels still read at ~60 GB/s effective against 360 with occupancy, coalescing, vec-

Date	Decision	Rationale
	moved to four-rows-per-workgroup -- one activation load dotted against four rows' weights -- and every dispatch grid to ceil(rows/4). Gain measured, small: 19.8 -> 20.3 tok/s end-to-end, 36.6 -> 35.3 ms marginal. Two negative results recorded with it: the x-traffic hypothesis behind NR4 was mostly wrong (L2 already covered the shared activations), and a LOOM_NO_BARRIER diagnostic (garbage output, valid timing) bounded all ~540 per-token barrier drains at ~7% -- barriers are not the bottleneck either. README gains a benchmarks section documenting both GPU chains step by step	torized loads, submission count and barriers all ruled out by measurement; the remaining gap needs real GPU profiling (Nsight) rather than another remote hypothesis. Negative results are logged because the next person will otherwise re-run these exact experiments. 89/89 on the 3060 and llvmpipe throughout
2026-07-31	**First real-GPU validation** (rented RTX 3060 12 GB, driver 580.173, Ubuntu 22.04): all 85 tests pass on NVIDIA silicon unmodified, and DeepSeek-Coder-V2-Lite Q4_K_M generation is token-identical to the CPU path. Then the two predicted fixes, measured: bring-up shape 0.6 tok/s (weights in host memory, ~500 submit-and-waits per token, 131 MiB VRAM in use); `allocDevice` + staged `upload` puts weights in VRAM and the routed-MoE chain becomes one recorded submission (`beginCmd`/`record`/`barrier`/`submitWait`) -> 1.5; a measured matvec size cutover (small projections to the CPU, fused MoE + 172 MB lm_head to the GPU) -> **4.3 tok/s vs 3.1 CPU-only**. Cutover default 2 MB (`LOOM_VK_MIN_BYTES` overrides); sweep was flat across 2-32 MB because no matvec sits between 3.6 MB and 172 MB	The discrete-GPU lesson in one number: a ~1 ms submit-and-wait makes a 3.6 MB matvec a loss on a device with 15x the CPU's bandwidth. The Vulkan backend still runs attention, norms and rope on the host, so every layer round-trips PCIe; Metal's 44 tok/s came precisely from eliminating those round trips (1.8 command buffers per token). The path to NVIDIA parity and beyond -- the 3060's 360 GB/s is ~2.4x the M5's unified memory, a 3090's 936 GB/s ~6x -- is the same port Metal already proved: attention kernels, then layer-tail, then whole-token recording. Fixed setup note: Ubuntu's unattended-upgrade replaced the NVIDIA userspace under the loaded module on first boot (Vulkan then enumerates only llvmpipe); one reboot fixes it, and the auto-update timers are disabled on the box so it cannot recur mid-benchmark
2026-07-30	`moeFfnBlockRouted`: a whole routed MoE layer -- route, id-indexed gate/up, per-slot SwiGLU, id-indexed down, device-gated reduce, shared expert -- as one command buffer, taking router *logits*. Local path only; the distributed path's experts are store offsets, not tensor planes, and keep host routing. Correct by a differential test against the host-routed block and by identical generated text	First measured at **0.2 tok/s against the host's 3.7** -- and the cause was the session's recurring bug, fourth instance: `gguf run` has its own generation loops and never called `materializeArenas`, so the mappings `deepseek.load` registers were never wired and every id dispatch paid GPU page faults on file-backed memory (~285 MB/s; 33-107 ms for dispatches that cost ~0.7 ms resident). Found by splitting the fused buffer into per-phase submissions and timing each (`LOOM_FUSED_DEBUG`, kept as the instrument). With one `materializeArenas` after each `parallelBegin` the order flips: fused 1.7 tok/s against host 0.4, back to back, identical text -- the first GPU path to beat the host on this model. Absolute numbers on the develop-

Date	Decision	Rationale
		ment machine remain unstable; the per-phase timings and the same-minute relative result are the evidence
2026-07-30	The default `--ram-gb 4.0` is machine-independent and OOM-killed the origin on an 8 GB box (7.4 GB anonymous RSS); `--ram-gb 0.5` runs stably at 0.52-1.39 GB	The RAM tier is sized by a constant rather than by what the machine has, and for an origin serving from a complete local copy it is pure waste -- every shard is local, so the cache can never hit. A default that kills the process on the class of hardware the project targets is the wrong default
2026-07-29	Steady-state decode on DeepSeek-V2-Lite Q4_K_M is 128 ms/token (7.8 tok/s) on a 17 GB M5 with the 9.7 GB store device-resident: expert FFN 58.9 ms (46%), expert get 26.3 (21%), attention 17.0 (13%), submissions 6.9 (5%)	Against ~2.5 tok/s at the start of the same session, from two changes that were both about reaching the weights rather than about arithmetic: making the mapping device-resident, and adding the Q5_0/Q8_0 kernels without which every MoE layer declined to the host. The expert FFN moves ~1.1 GB per token in 58.9 ms, about 19 GB/s against a ~110 GB/s streaming ceiling -- the gap is random 6 MB reads with dequantization, not the kernel, and it is where any further work belongs. No comparison against llama.cpp is claimed: the harness used earlier timed from first stream delta to last, which collapses when a server buffers its output, and it once reported 1236 tok/s. Those figures are withdrawn
2026-07-28	Batched prefill (`stepBatch`, `ggml.matmul`) and vectorized attention inner loops	Decoding unpacks every weight once per token, but a prompt is known up front, so one unpacked weight can serve several tokens. Measured 1.4x on a 145-token prefill; the kernel microbenchmark shows 2.4x in isolation, and the gap is attention, which is quadratic in prompt length and bound by the KV cache rather than by weight reads, so there is nothing to amortize. Vectorizing the attention dot and value-accumulation loops was worth a further 11% on decode. Batch capped at 8: past that, register pressure costs more than the sharing returns

Appendix B: Source layout

```
spec/          protocol specification (SPEC.md)
docs/          roadmap and planning
whitepaper/    this document
src/main.zig   CLI entry
src/core/      hashing/Merkle, tensor math, int4 quant, stats, iobench
src/engine/    loom-format MoE engine (MLA, router, expert cache, checkpoints)
src/gguf/      GGUF parser, GGML + codebook kernels, MLA and GQA engines, shared MoE routing, BPE
src/p2p/       distribution: wire frames, gossip, committees, sync, token-loop fetch
src/node/      daemon: node orchestration, RPC, model resolver, light node, metering
```

Appendix C: Provenance

Loom's design draws on the colibri expert-streaming technique; llama.cpp and GGML for the GGUF format, quantization layouts, and reference kernel semantics [7]; the DeepSeek V2/V3 model architecture (MLA, sparse routing) [1][2]; Ethereum networking patterns (bootnodes, ENR, gossip, committees) [12]; and content-addressed storage (Merkle verification, version-pinned manifests) [10]. The implementation is original Zig with three source dependencies, all compiled into the static binaries: blockblaz/zig-snappy [13] (wire compression), google/brotli v1.1.0 (at-rest chunk text), and zigstack/vector-index (vector similarity search, extracted from this project). FAISS, when present on a node, is loaded at runtime by vector-index and never required.